# Blender Lessons — Mill Alley Track

### 75 daily lessons, Oct 19, 2026 to Jan 31, 2027

## How to use these lessons

Each workday (Monday, Tuesday, Friday, Saturday and Sunday) has one lesson that teaches exactly what that day's Mill Alley task needs. Do it before the Mill Alley task, in this order:

1. **Watch and read first.** Start with the video or article, then keep the Blender Manual page open while you work.
2. **Steps.** Follow them in Blender 5.2 LTS with the menus and shortcuts given. Menu paths are written as *Menu > Submenu > Item*.
3. **Common mistakes.** Read these before the drill; they're the problems you're most likely to hit.
4. **Drill.** Do it from a blank file, timed. It's only done when every point under *Pass when* is true. If it isn't, redo it on the next buffer day.
5. **Use it on Mill Alley.** Then do the day's Mill Alley task with what you just practised.

Wednesdays and Thursdays are off. Buffer days have no new lesson: use them to catch up or redo drills. Tick lessons off in the tracker app, where every lesson also opens in full.

# Week 1 · Oct 19 to 25 · Interface, modeling basics, camera match

This week's goal is to move around Blender without thinking about it, model simple shapes to exact sizes, match a camera to the reference photo, and confirm that what you build in Blender arrives in Unreal Engine 5 at the right scale.

## Mon Oct 19 · Install, interface, navigation and transforms

**Goal:** install Blender 5.2 LTS, set it up for game work, and learn to move, select and transform objects.

**Time:** about 60 minutes for the lesson and 15 for the drill, then the Mill Alley task.

### Watch and read first

- **Video:** Blender 5 Complete Beginners Guide - FREE COURSE - Part 1 - The Interface — Grant Abbitt (Gabbitt) · Whole video, 12:42 (Blender 5): interface, navigation, views, G/R/S, toolbar
- **Video:** Blender Donut Tutorial Part 1 (2026) — Blender Guru · Optional second pass, 0:00-15:42 (install, navigation, adding and moving objects); Blender 5.0
- **Manual:** Navigation — Orbit, pan, zoom, numpad views; skim, 10 min
- **Manual:** Transform: Move — G/R/S with axis locks and typed values; Rotate and Scale pages sit next to it

### Steps

1. **Install.** Download Blender 5.2 LTS from blender.org/download (or Steam). Use the LTS version all project long so menus match these lessons.
2. **Set preferences** (Edit > Preferences). Under *Input*, tick *Emulate 3 Button Mouse* only if you have no middle mouse button, and *Emulate Numpad* if your keyboard has no number pad. Under *Interface*, leave everything default for now. Preferences save automatically.
3. **Learn the layout.** The big area is the 3D Viewport. Top right is the Outliner (a list of everything in the scene); below it is the Properties editor (tabs for render, output, scene, world, object, modifiers, materials and more). The tabs along the top (Layout, Modeling, Sculpting, UV Editing, Shading…) are workspaces: preset layouts for each job. Stay in Layout today.
4. **Navigate.** Middle-mouse drag orbits; Shift + middle-mouse pans; the scroll wheel zooms. Numpad 1, 3 and 7 give front, right and top views; hold Ctrl for the opposite side. Numpad 5 toggles perspective and orthographic. Numpad . (period) frames the selected object; Home frames everything. The ` (backtick) key opens a pie menu of all views. You can also drag the gizmo in the top right of the viewport.
5. **Select.** Left-click selects; Shift-click adds to or removes from the selection; A selects all; Alt+A deselects all; B box-selects; C paints a selection circle (right-click or Esc to stop).
6. **Add and delete.** Shift+A > Mesh > Cube adds a cube at the 3D cursor. X (or Delete) deletes. Shift+D duplicates and starts moving the copy.
7. **Transform with G, R and S.** Press G to move, R to rotate, S to scale. While moving, press X, Y or Z to lock to that axis (Shift+Z locks to the other two), type a number, and press Enter. Example: G Z 1 Enter moves the object up 1 m; R Z 90 Enter turns it 90 degrees; S 0.5 Enter halves it. Right-click or Esc cancels. Alt+G, Alt+R and Alt+S reset location, rotation and scale.
8. **Use the Adjust Last Operation panel.** After most actions a small panel appears at the bottom left of the viewport. Open it to type exact values after the fact. F9 brings it back.
9. **Save often.** Ctrl+S saves. Blender keeps the previous save as a `.blend1` backup next to your file. For versions, use File > Save As and click the + button beside the file name to add a number (alley_v01, alley_v02…).

### Common mistakes

- Moving the mouse outside the window while transforming. Watch the header text: it shows the axis and value you are typing.
- Scaling in Object mode and forgetting it. Object scale that isn't 1.0 will distort bevels and modifiers later. You'll learn to apply it tomorrow.
- Losing an object. Click it in the Outliner, then press Numpad . to frame it.

### Drill: table and chair from cubes (15 minutes)

The default cube is 2 m on each side. Build a table about 120 × 80 cm with its top 75 cm off the floor, four legs, and a chair whose seat is 45 cm high. Use only cubes, G, R, S and typed numbers (for example, S Z 0.02 makes the 2 m cube 4 cm tall).

**Pass when:** both objects stand on the grid floor (nothing below Z 0), the table top is roughly 75 cm high (check with N > Item > Dimensions), and it took you under 15 minutes without hunting for menus.

### Use it on Mill Alley

Today's Mill Alley task is the reference board. If you build it in Blender, drag each image into the viewport or use Shift+A > Image > Reference, and arrange them in rows by zone (foreground, left wall, right wall, midground, background). Save the file as `MillAlley_RefBoard.blend`.

## Tue Oct 20 · Edit Mode: extrude, inset, loop cut and bevel

**Goal:** edit the mesh itself (its vertices, edges and faces) with the four tools you'll use on every kit piece.

**Time:** about 60 minutes for the lesson and 20 for the drill.

### Watch and read first

- **Video:** Blender 5 Complete Beginners Guide - FREE COURSE - Part 4 - Old Man Details — Grant Abbitt (Gabbitt) · 1:31-13:25: Edit Mode, vertex/edge/face modes, Extrude, Loop Cut, Inset, face loops (~12 min). Bevel is not covered here; see the manual Bevel page
- **Manual:** Mesh Structure — Vertices, edges, faces, loops; 10 min read
- **Manual:** Loop Cut — Ctrl+R; neighbours in the same menu: Extrude Faces, Inset Faces, Bevel

### Steps

1. **Object Mode vs Edit Mode.** Object Mode moves whole objects. Tab switches the selected object into Edit Mode, where you change its shape. The mode menu at the top left of the viewport shows which mode you're in.
2. **Vertices, edges, faces.** In Edit Mode press 1, 2 or 3 on the top number row for vertex, edge or face selection. G, R and S work on the selection exactly as they did on objects.
3. **Useful selection.** Alt+click an edge to select its whole loop. L selects everything linked to the face under the mouse. Ctrl+Numpad + grows the selection. H hides the selection and Alt+H brings it back, which helps you reach the inside of things.
4. **Apply scale first.** In Object Mode, Ctrl+A > Scale sets the object's scale back to 1.0 while keeping its size. Do this after you scale an object and before you bevel it, or the bevels will be uneven.
5. **Extrude (E).** Select a face and press E to pull out new geometry along its normal. Type a value: E 0.1 pulls out 10 cm; E −0.1 pushes in. Right-click during the extrude to leave the copy in place, then move it yourself.
6. **Inset (I).** Select a face and press I to make a smaller face inside it, joined by a border. Type the border width, for example I 0.08. Pressing I a second time while insetting switches between one inset for the whole selection and one per face.

7. **Loop Cut (Ctrl+R).** Hover over an edge: a yellow preview loop appears across it. Scroll to add more cuts, left-click to confirm, then slide it or right-click to keep it centered.
8. **Bevel (Ctrl+B).** Select edges and press Ctrl+B, move the mouse to set the width, and scroll to add segments. Type the width directly, e.g. Ctrl+B 0.01 for 1 cm. Ctrl+Shift+B bevels vertices instead.
9. **Clean-up keys.** X opens the delete menu (vertices, edges, faces, dissolve). M merges vertices. F fills the selection with a face. Ctrl+Z undoes; Ctrl+Shift+Z redoes.

### Common mistakes

- Extruding and then cancelling with Esc. The extrude is cancelled but the duplicate geometry stays, stacked on the original. Press Ctrl+Z, or select everything and use Mesh > Clean Up > Merge by Distance.
- Bevelling on an object whose scale isn't applied. Apply scale first (step 4).
- Loop cuts that stop halfway. They stop at triangles and n-gons (faces with more than four sides), which is a sign your topology needs cleaning.

### Drill: doorframe with a recessed door (20 minutes)

1. Add a cube. S Z 1.5 Enter, then G Z 1.5 Enter, then S Y 0.15 Enter. You now have a wall 2 m wide, 3 m tall and 30 cm deep, standing on the floor. Ctrl+A > Scale.
2. Tab into Edit Mode and press Numpad 1 for the front view. Ctrl+R over a top edge, scroll to 2 cuts, click, then right-click. In edge mode, Alt+click one of the new vertical loops, Shift+Alt+click the other, and press S X 1.5 Enter to move them out to ±50 cm, giving a 1 m opening.
3. Ctrl+R over a vertical edge, click, right-click, then G Z 0.6 Enter to move the cut to 2.1 m high.
4. In face mode, select the front face in the lower middle (the door area). I 0.08 Enter makes the frame. Then E −0.12 Enter pushes the door back 12 cm.
5. Select the edges around the door opening and Ctrl+B 0.01 with 2 segments.

**Pass when:** the opening is 1.0 × 2.1 m (measure with N > Item while vertices are selected, or with the Measure tool), the door sits back from the frame, nothing has stacked duplicate faces (Mesh > Clean Up > Merge by Distance reports 0 removed), and N > Item shows scale 1, 1, 1.

### Use it on Mill Alley

Today you paint over the reference and circle every repeating element. For each kit piece you circle (A01 wall, A03 arch, B03 window and so on), think about how you'd make it with today's four tools: a wall with an inset, extruded window recess and bevelled sill. Note this next to each ID on the paintover.

## Fri Oct 23 · Cameras and matching the reference photo

**Goal:** set up a Blender camera whose view lines up with the reference photo, so everything you model can be checked against it.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Camera Perspective Matching in Blender! (Free & Paid Tools) — Nik Kottmann · 6:08-9:37 for the free fSpy workflow (2022, but the fSpy steps are unchanged); 0:33-6:08 shows the paid Perspective Plotter add-on, optional
- **Read:** fSpy-Blender importer v1.0.3 (GitHub releases) — stuffmatic · Install the release zip via Preferences > Add-ons > Install from Disk (do not unzip). Last release 2019 but users report it works in Blender 5.0/5.1 (issue #90); add a camera to the scene before importing
- **Read:** FTC - Fit The Camera (extension) — TonyS87 · In-Blender alternative to fSpy: vanishing-point camera match; v1.1.1 (Oct 15 2026), requires Blender 5.2 LTS
- **Manual:** Cameras — Lens (Focal Length, Shift), Camera (Sensor Fit, Size) and Background Images sections

## Steps

1. **Match the frame shape.** In Output Properties > Format, set Resolution X and Y to the photo's pixel size (or the same ratio, e.g. 3000 × 4000 → 1500 × 2000). If the shape is wrong, the photo can never line up.
2. **Add a camera.** Shift+A > Camera. With the camera selected, Ctrl+Numpad 0 makes it the active camera and looks through it. Numpad 0 looks through the active camera later.
3. **Load the photo.** Select the camera, open its Object Data properties (the green camera icon), tick *Background Images*, click *Add Image* and open the photo. Set Opacity to about 0.6. If the photo hides your geometry, set Depth to *Front* while checking alignment.
4. **Set the lens.** In the same tab under *Lens*, set Focal Length. If the photo has EXIF data (right-click > Properties > Details on Windows), use its 35 mm-equivalent focal length with Sensor Fit *Auto* and Sensor Size 36 mm. Phone wide cameras are about 24–26 mm. If you don't know, start at 24 mm.
5. **Build guides.** Add a large plane for the ground. Add two long boxes for the two walls, the width apart that the alley is (from the plan's 1:1 scale table). These give you lines to match against the photo's vanishing lines.
6. **Place the camera.** Set the camera's location Z to eye height, about 1.6 m, and rotation X to 90° so it looks level. A level camera keeps vertical lines vertical, as in the photo. If the photo's verticals lean, tilt the camera a few degrees instead.
7. **Move the camera into place.** In the viewport sidebar (N > View) tick *Lock Camera to View*. Now orbiting and panning in camera view moves the camera. Line up the wall boxes' bottom and top edges with the photo's brick courses and ground line. Untick it when you're done so you don't nudge the camera by accident.
8. **Fine tune.** Change the focal length and move the camera back or forward together: a longer lens further back flattens the perspective; a shorter lens closer in exaggerates it. If the photo was cropped off-centre, use Shift X and Shift Y under *Lens* to move the frame without rotating.
9. **Optional: fSpy.** The free fSpy app works out a camera from vanishing lines you draw on the photo. Its Blender importer (v1.0.3 on GitHub, installed from the zip) still works in Blender 5.x according to users. The *Fit The Camera* extension (Get Extensions) does a similar job inside Blender. Matching by hand as above teaches you more; use these to check your result.
10. **Lock it.** Name the camera `CAM_01_Reference`, then in the Properties > Object tab click the lock icons (Transform > Location, Rotation) so it can't be moved.

## Common mistakes

- A wrong render aspect ratio. Fix step 1 before anything else.
- Matching with the camera tilted when the photo is level. If the photo's vertical lines are straight up and down, keep rotation X at exactly 90°.
- Changing only the focal length. Focal length and distance work as a pair. Match the vanishing point first, then the lens.

## Drill: match a photo of your own (30 minutes)

Take a photo of a hallway, street or room where you can measure one thing (a door is about 2.1 m tall; a ceiling is about 2.4 m). Match a camera to it using steps 1 to 8 with a box for the measured object.

**Pass when:** the box's edges sit on the photo's edges within a few pixels at the near end and the far end, and the box is the real measured size.

## Use it on Mill Alley

Today's task is the Mill Alley camera match. Use the plan's scale breakdown for the alley width and the floor heights (400 cm left, 380 cm right). Block the two buildings as boxes at those sizes, match the camera, and save the file as

`MillAlley_CameraMatch.blend`. This camera becomes Camera 1 for every review and the final beauty shot.

## Sat Oct 24 · Units, snapping, origins, naming and FBX export to UE5

**Goal:** lock the settings that make modular pieces line up and import into UE5 at the right size.

**Time:** about 75 minutes for the lesson and 20 for the drill.

### Watch and read first

- **Video:** Blender 4.5 to Unreal Engine 5.6 - Fix Scale & Animation in 2025 — Imphenzia · Watch the static-mesh/scale part; skip the animation section
- **Manual:** Scene Properties (Units) — Units panel: Unit System, Unit Scale, Length
- **Manual:** Snapping — Increment and Absolute Grid Snap; also see 3D Cursor, Object Origin, Collections
- **Manual:** FBX (Legacy) — In Blender 5.2 this legacy page documents the FBX exporter; the main FBX page now covers the new importer. Some options aren't written up yet
- **Unreal docs:** Importing Static Meshes in Unreal Engine (UE 5.8 docs) — States 1 Unreal unit = 1 cm; import steps
- **Unreal docs:** Units of Measurement in Unreal Engine (UE 5.8 docs) — Distance/length default unit is centimetres

### Steps

1. **Units.** Scene Properties > Units: Unit System *Metric*, Unit Scale *1.0*, Length *Meters*. One Blender unit is now one metre. Leave it this way for the whole project.
2. **Grid.** Viewport Overlays (the two-circles dropdown) > Guides > Scale: set to 0.25 so the grid lines are 25 cm apart, the project's snap size.
3. **Snapping.** Click the magnet dropdown in the viewport header. Set *Snap Base* to Closest and *Snap Target* to Increment, and tick *Absolute Grid Snap*. Turn snapping on with the magnet (Shift+Tab) or hold Ctrl while moving to snap temporarily. Typed values are always exact: G X 0.25 Enter moves 25 cm.
4. **Vertex snapping.** In the same menu, also tick *Vertex*. Now while moving with Ctrl held, a piece's corner snaps to another piece's corner. This is how modules meet with no gap.
5. **The 3D cursor.** Shift+right-click places the 3D cursor; new objects appear there. Shift+S opens the snap pie: *Cursor to World Origin*, *Cursor to Selected*, *Selection to Cursor*.
6. **Origins (pivots).** The orange dot is the object's origin, which becomes the pivot in UE5. The project rule is bottom-left-back. In Edit Mode, select that corner vertex, press Shift+S > *Cursor to Selected*, Tab back to Object Mode, then Object > Set Origin > *Origin to 3D Cursor*. Then Alt+G moves the object so its origin sits at the world origin.
7. **Collections.** Press M with objects selected to move them into a new or existing collection. Keep one collection per module family (Kit_A, Kit_B…). Rename with F2 or by double-clicking in the Outliner.
8. **Naming.** Use the plan's convention: `SM_MillAlley_<Family>_<Piece>_<W>x<H>`, e.g. `SM_MillAlley_A_WallFlat_400x400`. UE5 uses the object name as the asset name.
9. **FBX export.** Select the object(s), then File > Export > FBX (.fbx). Blender 5.2 has a new FBX importer, and the exporter may be listed as *FBX (legacy)*; it's the same exporter with the same options. Tick *Limit to: Selected Objects*; Object Types: *Mesh* only. Under Transform, set *Apply Scalings* to **FBX Units Scale** and leave Forward and Up at their defaults. Under Geometry, set *Smoothing* to *Face* and keep *Apply Modifiers* on. Save these settings as a preset with the + next to *Operator Presets* (name it `UE5_StaticMesh`).
10. **Import in UE5.** Drag the FBX into the Content Browser. Keep the default static mesh options and import. Open the mesh: the viewport text shows *Approx Size*. Drop it into the level and check the Details panel scale is 1, 1, 1.

### Common mistakes

- Importing at 1/100 size or 100× size. The cause is almost always *Apply Scalings*. With metres in Blender and *FBX Units Scale*, a 1 m cube arrives as 100 cm.
- Pivots in the middle of modules. UE5 snaps by pivot, so a centred pivot breaks the 25 cm grid.
- Exporting unapplied scale. Apply scale (Ctrl+A > Scale) before export so UE5 sees scale 1.

### Drill: the 4 × 4 m wall round trip (20 minutes)

Make a wall block 4 m wide, 4 m tall and 30 cm deep with the origin at its bottom-left-back corner, named `SM_Test_Wall_400x400`. Export it with your preset and import it into UE5.

**Pass when:** UE5 shows Approx Size 400 × 30 × 400 (axis order may differ), the scale in the level is 1, 1, 1, and dragging it with UE5's grid snap set to 25 lets two copies meet edge to edge with no gap.

### Use it on Mill Alley

Today you lock the metrics and naming and set up the UE5 project. Save your export preset, write the units, grid, pivot and naming rules into the metrics sheet, and commit the empty UE5 project plus the test wall to source control.

## Sun Oct 25 · Viewport shading and a quick overcast light

**Goal:** light a greybox simply so you can judge forms and mood, and produce the base for the mood paintover.

**Time:** about 45 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Blender 5 Complete Beginners Guide - FREE COURSE - Part 2 Materials — Grant Abbitt (Gabbitt) · 4:30-11:09: camera view, viewport modes, adding lights, render engines, rendering an image
- **Video:** Blender 5 Complete Beginners Guide - FREE COURSE - Part 8 - Lights & Composit — Grant Abbitt (Gabbitt) · Optional, 6:30-9:15: World shader and area light
- **Manual:** Viewport Shading — Solid / Material Preview / Rendered modes
- **Manual:** Light Objects — Sun light settings; World colour

### Steps

1. **Shading modes.** Z opens the shading pie: Wireframe, Solid, Material Preview and Rendered. Solid is for modeling; Rendered shows real lights.
2. **Solid mode options.** Click the dropdown next to the shading buttons. Lighting *Studio* or *MatCap*; Color *Object* or *Random* (tells pieces apart); tick *Cavity* to see edges and *Shadow* for a quick sense of depth.
3. **Pick EEVEE.** Render Properties > Render Engine: *EEVEE*. It's fast and close enough for lighting studies. (Cycles is slower but physically accurate; you'll use it for baking.)
4. **World light.** World Properties > Surface: set Color to a cool grey-blue and Strength to about 1. An overcast sky lights everything from above and all around, which is most of the photo's light.
5. **Optional HDRI.** For a better sky, download an overcast HDRI from Poly Haven (CC0), click the yellow dot next to Color, choose *Environment Texture*, and open the file.
6. **A weak sun.** Shift+A > Light > Sun. Set Strength to about 1–2 and Angle to 15–30° so the shadows are soft. Rotate it so light comes from the far bright opening, like the photo. Make its colour slightly warmer than the sky.
7. **Exposure.** Render Properties > Color Management: View Transform *AgX* (the default), Look *None*, and use *Exposure* to brighten or darken the whole image without touching lights.
8. **Render.** Look through Camera 1 (Numpad 0), press F12, then Image > Save As in the render window. Save it as a PNG next to the reference.

## Common mistakes

- A bright sun and black shadows. An overcast scene is lit mostly by the sky; keep the sun weak and soft.
- Judging lighting in Material Preview. It uses its own built-in light; switch to Rendered to see yours.
- Fixing brightness with light strength. Use Exposure for overall brightness and lights for direction and contrast.

## Drill: overcast greybox frame (30 minutes)

Use your camera-match file. Give the walls and ground a plain grey material (Material Properties > New, Base Color mid-grey), set the world and sun from steps 4 to 7, and render Camera 1.

**Pass when:** the far end of the alley is the brightest area, shadows are soft-edged, the two walls and the ground have clearly different values, and nothing is pure black or pure white.

## Use it on Mill Alley

This render is the base for today's mood paintover. Paint over it (in any image editor) toward the photo: cool, wet and overcast. Save it beside the reference as the target image. For the standard check, put your reference board and one composition reference (a film still or painting with a similar long, framed view) together, as the Farm and Persian projects do; that's Review 1.

# Week 2 · Oct 26 to Nov 1 · Blockout modeling

This week's goal is to build fast, accurate blockouts with modifiers so the greybox can change without being rebuilt, and to move meshes between Blender and UE5 without breaking anything.

## Mon Oct 26 · Modifiers: Array, Mirror and Solidify

**Goal:** use modifiers to repeat, mirror and thicken shapes non-destructively, meaning you can change the original at any time.

**Time:** about 60 minutes for the lesson and 25 for the drill.

### Watch and read first

- **Video:** Modifiers for Beginners (Blender Tutorial) — Ryan King Art · 1:42-13:40 Mirror, 24:55-28:42 Array, 34:47-36:09 Solidify, 41:20-45:39 apply scale and stack order (~22 min total)
- **Manual:** Array Modifier — Relative/constant offset, count
- **Manual:** Solidify Modifier — Also Mirror Modifier; stack basics: manual page

### Steps

1. **What a modifier is.** A modifier changes how a mesh looks without changing the mesh underneath. Open Properties > Modifiers (the blue wrench) and click *Add Modifier*. The menu is grouped into Edit, Generate, Deform, Normals and Physics.
2. **Array (Generate > Array).** It repeats the mesh. *Count* sets how many copies. *Relative Offset* moves each copy by a multiple of the object's own size (Factor X 1 = side by side). *Constant Offset* moves each copy by an exact distance, e.g. Z 4 m for one storey. Tick *Merge* to weld touching copies.
3. **Stack two arrays.** Add a second Array below the first: the first makes a row of windows along X, the second repeats that row up Z. Modifiers run top to bottom, so order matters. Drag them by the handle at the top right of each panel.
4. **Mirror (Generate > Mirror).** It mirrors the mesh across the object's origin. Pick the axis, tick *Clipping* so vertices on the centre line can't cross it, and tick *Bisect* to cut away anything already over the line. Model half and get the other half free.
5. **Solidify (Generate > Solidify).** It gives a flat surface thickness. Set Thickness (e.g. 0.3 for a 30 cm wall) and Offset −1 or 1 to grow inward or outward from the surface.
6. **Show and hide.** The icons in each modifier header toggle it in Edit Mode, in the viewport and in renders. Hide a modifier to see the base mesh again.
7. **Apply only when you must.** Hover a modifier and press Ctrl+A (or use its dropdown > Apply) to make it real geometry. Keep modifiers live during blockout; the FBX exporter applies them for you when *Apply Modifiers* is ticked.

### Common mistakes

- An Array that shoots copies far away. Unapplied scale or rotation multiplies the offset; apply scale and rotation (Ctrl+A) first.
- A Mirror that mirrors around the wrong place. It uses the object's origin; move the origin or set *Mirror Object* to an empty.
- Applying modifiers too early. Once applied, you lose the quick edits that make blockouts fast.

### Drill: a 5-storey wall with arrayed windows (25 minutes)

1. Make a plane 4 m wide and 20 m tall standing up (S X 2, S Y 10, then R X 90, or use N > Item > Dimensions). Add Solidify, Thickness 0.3.

2. Make a separate box 1.4 × 0.4 × 2.2 m (a window proxy). Place it 90 cm above the floor, sticking through the wall.
3. Give the window proxy an Array (Count 2, Constant Offset X 2.0 m, Relative Offset off) and a second Array (Count 5, Constant Offset Z 4.0 m).

**Pass when:** 10 windows sit at even 4 m storey heights on a 20 m wall, and changing the first window's width updates all 10.

### Use it on Mill Alley

Today you greybox the ground and both building masses in UE5 at scale. Make the building masses in Blender with Solidify on simple planes and arrays for the floors, then export them. Use the plan's sizes: about 5 m wall to wall, about 20 m tall, 400 cm floors on the left and 380 cm on the right.

## Tue Oct 27 · Boolean openings, Spin for arches, and the Bevel modifier

**Goal:** cut openings with booleans, make round arch heads with Spin, and soften edges with the Bevel modifier.

**Time:** about 70 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Modifiers for Beginners (Blender Tutorial) — Ryan King Art · 18:49-24:55 Bevel modifier, 30:53-33:29 Boolean (~9 min)
- **Video:** How to Use the Spin Tool in Blender — Ryan King Art · Whole video (Jan 2026); use it for the round arch head
- **Manual:** Boolean Modifier — Difference cutter workflow; Exact vs Fast solver
- **Manual:** Spin — For the round arch head; Bevel Modifier

### Steps

1. **Make a cutter.** A cutter is a separate object shaped like the hole you want. In Object Properties > Viewport Display, set *Display As* to *Wire* so you can see through it, and turn off its render visibility (camera icon in the Outliner).
2. **Boolean modifier.** Select the wall, Add Modifier > Generate > Boolean. Operation *Difference*, Operand Type *Object*, pick the cutter. Use the *Exact* solver: it is slower but handles overlapping faces. For several cutters, put them in one collection and set Operand Type to *Collection*.
3. **Keep cutters live.** Moving or scaling the cutter moves the hole. This is perfect for blockouts. Before final modeling you'll rebuild openings with clean geometry (Week 4).
4. **Round arch cutter.** Add a Cylinder (in the Adjust Last Operation panel set Vertices 32, Radius 1.5, Depth 1) and rotate it R X 90. Add a box below it for the straight jambs and join the two with Ctrl+J. That makes one cutter for a 3 m wide round-headed opening.
5. **Spin.** Spin sweeps selected geometry around the 3D cursor, seen from the current view. In Edit Mode, select the cross-section of an arch moulding (for example one edge 30 cm long), place the 3D cursor at the arch's centre (select a vertex there, Shift+S > *Cursor to Selected*), go to front view (Numpad 1), and pick the *Spin* tool in the toolbar on the left. Drag its plus-shaped handle, then in the Adjust Last Operation panel set Angle 180° and Steps 16.
6. **Bevel modifier.** Add Modifier > Generate > Bevel. Amount 0.01–0.02 (1–2 cm), Segments 2, Limit Method *Angle* (30°), so only sharp edges are bevelled. Small bevels catch light and stop blockouts looking like paper.
7. **Order.** Put Boolean above Bevel so the bevel also rounds the edges of the hole.

### Common mistakes

- The hole doesn't appear. The cutter must pass fully through the wall; make it deeper than the wall.

- Strange shading around boolean holes. That's long, thin triangles. It's fine in a blockout; final pieces get clean topology.
- Spin going the wrong way or around the wrong point. Check the 3D cursor position and the view you're in: Spin uses both.

### Drill: the A03 blind arch recess as a proxy (30 minutes)

Model a wall 3 m wide and 6 m tall (A03 is 300 × 600 cm). Cut a round-headed recess 25 cm deep (not all the way through: a blind arch), with its arch 2.4 m wide and the top of the curve at 5.6 m. Then use Spin to add a brick arch ring 15 cm deep around the curve, and add a Bevel modifier.

**Pass when:** the recess is 25 cm deep, the arch is a clean half-circle seen from the front (Numpad 1), the arch ring follows it exactly, and the cutter still moves the recess when you move it.

### Use it on Mill Alley

Today you greybox the left facade with placeholders A01 to A09. Use your A03 proxy and booleans for the tall blind arches and the A05 loading door, with arrays to repeat them along the wall.

## Fri Oct 30 · Precise modeling: dimensions, typed values and measuring

**Goal:** model to exact centimetres every time, which is what makes a modular kit work.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Exact measurements - Blender 4 — Plockis · 0:00-3:00 (units, typed lengths, snapping to exact values); 3:00 onward uses the optional Edit Mesh Tools add-on. Whole video 6:09
- **Manual:** Sidebar — Item tab: Location, Rotation, Scale, Dimensions
- **Manual:** Numeric Input — Typed transforms (S 2 Enter); also Measure tool and Snapping

### Steps

1. **The Item panel.** Press N to open the sidebar and click the *Item* tab. In Object Mode it shows Location, Rotation, Scale and *Dimensions*. Type exact sizes into Dimensions, then apply scale (Ctrl+A > Scale).
2. **Item panel in Edit Mode.** With vertices selected it shows their *Median* (average position). Toggle *Global* to see world positions and type a value to move them exactly, e.g. set Z to 0.9 to put a sill at 90 cm.
3. **Typed transforms.** Every G, R and S accepts numbers and simple maths. G Z 0.9 Enter; S X 1.4/2 Enter; G X −0.25 Enter. Use Tab while typing to switch to the next axis field.
4. **Snap one vertex to another.** In the snap menu, tick *Vertex*. Move with G, hold Ctrl, and hover a target vertex: the selection jumps exactly onto it. In Edit Mode, set *Snap Base* to *Active* to snap from the last-selected vertex.
5. **Measure tool.** The ruler icon in the toolbar. Drag between two points; hold Ctrl to snap to vertices. Click a measure line's middle and drag to measure angles. Delete a measure with X while hovering it.
6. **Edge lengths.** In Edit Mode, open the Overlays dropdown and, under Measurement, tick *Edge Length*. Selected edges now show their length. Use it to check openings.
7. **Loop cuts to exact positions.** After Ctrl+R, right-click to centre the loop, then use G Z (or G X) with a typed value to move it to the exact height.

### Common mistakes

- Reading local instead of global values. In Edit Mode, check the *Global* toggle in the Item panel.
- Scaling an object to size and never applying it. Dimensions look right but modifiers and exports behave oddly.

- Eyeballing “close enough”. A 2 cm error repeated over ten modules is a 20 cm gap.

### Drill: a 300 × 380 cm bay with a 140 × 220 cm opening (30 minutes)

Model a wall section 300 cm wide, 380 cm tall and 30 cm deep, with a window opening 140 cm wide and 220 cm tall, centred left to right, its bottom 90 cm above the floor. Add a stone sill 160 × 15 × 20 cm projecting 5 cm. Use only typed values and snapping.

**Pass when:** the Edge Length overlay shows 1.4 m and 2.2 m on the opening, the brick piers each side are 80 cm (half of the 160 cm between neighbouring windows), the sill top is at 90 cm, and the origin is at bottom-left-back.

### Use it on Mill Alley

Today you greybox the right facade with B01 to B09. Your drill is B02; array it along the wall and add the B05 ledge (50 cm tall, 60 cm deep) and the B06 light well volume (120 cm wide, 200 cm drop).

## Sat Oct 31 · Building fast from primitives and collection instances

**Goal:** block out complex hero shapes quickly from simple primitives, and reuse them as instances.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to use and have full control of your collection instances (Blender tutorial) — Hugo Antunes (Art with Blender) · Whole video (2021; collection instancing UI unchanged)
- **Manual:** Collection (Instancing) — Collection instances for repeated pieces
- **Manual:** Array Modifier — Plank arrays

### Steps

1. **Think in boxes.** Every hero shape starts as a few boxes at the right size. Block the biggest forms first, then medium, then small. Stop before details.
2. **Planks with Array.** Make one board (e.g. 15 × 2 × 250 cm). Add an Array along Z with Relative Offset 1.0 plus a small gap (Constant Offset Z 0.005). Count sets how many boards. That’s cladding in one object.
3. **Corbel steps.** For the chimney cap, make the shaft as a box (180 × 180 cm). Select its top face, extrude up a little (E 0.1), then scale it out (S 1.1) and extrude again: each extrude-and-scale makes one corbel step.
4. **Collection instances.** Put a finished blockout piece in its own collection. In Collection Properties > Instancing, set its offset (or put the 3D cursor at its base and use *Set Offset from Cursor*). Then Shift+A > Collection Instance adds a lightweight copy. Edit the original collection and every instance updates.
5. **Linked duplicates.** Alt+D duplicates an object that shares the same mesh data. Edit one in Edit Mode and all copies change. Shift+D makes an independent copy.
6. **Hide and isolate.** Numpad / shows only the selection (local view); press it again to return. Use it to work on one piece without the rest of the scene in the way.

### Common mistakes

- Detailing too early. Blockouts answer “is the size and placement right?” Details belong to later weeks.
- Instances you can’t edit. Edit the source collection, not the instance; or Object > Apply > Make Instances Real if you need a separate copy.
- Arrays of separate planks exported as one mesh by accident. That’s fine for blockout; the final skybridge gets real planks in Week 6.

### Drill: the skybridge proxy in 30 minutes

Build the skybridge from the plan's sizes: an enclosed tier 500 cm span × 300 cm tall × 250 cm deep with its underside 600 cm above the ground, a girder deck 40 cm deep under it, two window openings per side (booleans), and cladding boards made with an Array. Set a 30-minute timer.

**Pass when:** it's done in 30 minutes, the sizes match the plan's table, and from Camera 1 it sits over the photo's skybridge.

### Use it on Mill Alley

Today you greybox the skybridge, chimney (180 × 180 cm, top at about 28 m with a corbelled cap), end building and loading door, and set Camera 1 plus 3 alternate cameras. Use your drill as the skybridge greybox.

## Sun Nov 1 · The round trip: updating meshes in UE5 without breaking anything

**Goal:** change a mesh in Blender and update it in UE5 so every placed copy updates and nothing moves.

**Time:** about 40 minutes for the lesson and 20 for the drill.

### Watch and read first

- **Video:** UE5 Quick Tip 39: How to Get UE5 Static Mesh Asset Import Dialog Menu Back on Reimports — WorldofLevelDesign · Whole video, 6:03 (Dec 2025)
- **Read:** Send to Unreal docs (community-maintained fork) — Poly Hammer (fork of Epic BlenderTools) · Optional. Latest Send to Unreal 2.6.7 (Nov 2025) is tested on Blender 3.6/4.2 and UE 5.3/5.4 only; 5.x support unconfirmed, so batch FBX / Collection Exporters is the safe default
- **Manual:** Collections: Exporters — Collection Properties > Exporters; Export All; File > Export All Collections. FBX is listed as 'FBX legacy'
- **Manual:** FBX (Legacy) — Exporter options (Batch Mode, Forward/Up)
- **Unreal docs:** Importing Assets Using Interchange in Unreal Engine (UE 5.8 docs) — Reimport section: UE remembers the pipeline and options used
- **Unreal docs:** Reimporting Assets Automatically in Unreal Engine (UE 5.8 docs) — Auto-reimport when the source FBX changes; monitored folders

### Steps

1. **What UE5 remembers.** A placed mesh in a UE5 level points to an asset. If you reimport the asset, every placed copy updates. The placement stays right as long as the name and pivot stay the same.
2. **Re-export to the same file.** Export the changed object with the same name, preset and file path as before, overwriting the old FBX.
3. **Reimport in UE5.** In the Content Browser, right-click the asset > *Reimport*. Or turn on auto reimport in Editor Preferences > Loading & Saving so UE5 picks up changed files automatically.
4. **Batch export with Collection Exporters.** In Blender 4.2 and later, open Collection Properties for a collection, find *Exporters*, click *Add Exporter* > FBX (listed as *FBX legacy*), and set the file path and your export settings. File > Export All Collections then exports every collection that has an exporter in one click.
5. **Send to Unreal (skip it for now).** Epic's free add-on exports straight into an open UE5 project, but Epic's own release was last tested on Blender 3.6, and the community fork (poly-hammer) on Blender 4.2. Neither claims Blender 5.x support, so use Collection Exporters and plain FBX, which always work.
6. **Keep a scale figure.** Put a 180 cm figure in every blockout: UE5's Manny or Quinn mannequin, or a 180 cm capsule in Blender. Your eye judges scale from people, not numbers.

### Common mistakes

- Renaming the mesh before re-export. UE5 imports it as a new asset and your placed copies don't change.

- Moving the pivot. Every placed copy jumps by the same amount. Fix the pivot in Blender and reimport.
- Exporting with a different preset. Scale or axis changes break every placement.

## Drill: change a proxy and update it in UE5 (20 minutes)

Take one exported greybox piece, change it in Blender (add a floor, move a window), re-export to the same file and reimport in UE5.

**Pass when:** every placed copy in the level updates, nothing shifts position, and no duplicate asset appears in the Content Browser.

## Use it on Mill Alley

Today is blockout lighting, a walkthrough and Review 2; the greybox gets locked. Set up a Collection Exporter for each kit family now so later fixes are one click. For the standard check, place a 180 cm figure next to the blockout in your review screenshots, as Indian street and Le Chlanko do.

# Week 3 · Nov 2 to 8 · Materials, masks and baking

This week's goal is to build materials from free CC0 textures, add your own edge-wear and leak masks with nodes, bake them into game textures, and paint vertex colours that UE5 can use.

## Mon Nov 2 · The Shader Editor, PBR textures and a 2 m tile

**Goal:** load a full PBR texture set onto a material, understand each map, and scale it to the project's 200 cm tile.

**Time:** about 60 minutes for the lesson and 25 for the drill.

### Watch and read first

- **Video:** Blender Basic Texturing Tutorial (Donut Part 5) — Blender Guru · 0:20-15:57: image textures, what PBR textures are, importing a PBR set manually (~16 min; Dec 2025). Skip the Poliigon add-on section
- **Video:** Procedural Nodes For Beginners (Blender Tutorial) — Ryan King Art · 10:50-13:05 Node Wrangler shortcuts; 52:10-55:42 Vector nodes (Mapping)
- **Manual:** Principled BSDF — Base Color, Roughness, Metallic, Normal inputs
- **Manual:** Image Texture Node — Also Mapping Node; Node Wrangler; Non-Color

### Steps

1. **The Shading workspace.** Click the *Shading* tab. The bottom half is the Shader Editor, where materials are built as nodes connected left to right into the *Material Output*.
2. **A new material.** Select an object, go to Material Properties and click *New*. You get a *Principled BSDF* node: one shader with inputs for Base Color, Metallic, Roughness, Normal and more. It matches what UE5's materials expect.
3. **Turn on Node Wrangler.** Edit > Preferences > Add-ons, search "Node Wrangler", tick it. It ships with Blender.
4. **Download a texture set.** From ambientCG, pick a brick material and download the 2K PNG zip. Unzip it. You'll see Color, Roughness, NormalGL, NormalDX, AmbientOcclusion and Displacement maps.
5. **Load them in one go.** In the Shader Editor, click the Principled BSDF node, press Ctrl+Shift+T, select the Color, Roughness, NormalGL and Displacement files and click *Principled Texture Setup*. Node Wrangler wires them, adds a Normal Map node, and adds Texture Coordinate and Mapping nodes on the left.
6. **Color vs Non-Color.** Colour maps are *sRGB*. Every data map (roughness, normal, AO, height, masks) must be *Non-Color* in the Image Texture node's Color Space menu. Node Wrangler sets this; check it every time you wire a map by hand.
7. **OpenGL vs DirectX normals.** Blender uses OpenGL normals (NormalGL); UE5 uses DirectX (NormalDX). Use GL in Blender and DX in UE5, or tick *Flip Green Channel* on the texture in UE5.
8. **Scale to the 2 m tile.** Add a plane (it's 2 × 2 m with UVs covering 0 to 1). With Mapping Scale 1, the texture covers exactly 2 m. Count the brick courses: the project wants 24 per 2 m (8.33 cm each). If the texture shows 12, set the Mapping Scale to 2, 2, 2; if it shows 16, set 1.5. Always scale X and Y together.
9. **Preview.** Ctrl+Shift+click any node to preview it on the object (a Node Wrangler shortcut). Switch the viewport to Material Preview (Z) to see the material.

### Common mistakes

- Bumps going the wrong way. Usually a DX normal map in Blender or a Color-space normal map. Use NormalGL and Non-Color.
- Shiny, plastic-looking brick. The roughness map is set to sRGB, or plugged into the wrong input.
- Stretched bricks. X and Y were scaled differently; brick should be square-scaled.

### Drill: tile an ambientCG brick at exactly 2 × 2 m (25 minutes)

Make a 2 × 2 m plane, set up a brick texture set with Node Wrangler and set the Mapping scale so the plane shows exactly 24 brick courses.

**Pass when:** you count 24 courses top to bottom, the mortar lines look recessed, every data map is Non-Color, and a second plane placed edge to edge continues the pattern with no visible join.

### Use it on Mill Alley

Today you source the ten CC0 base sets (sooty brick, painted brick, setts, mud, concrete, moss, timber, rusted metal…) and rescale each to the 200 cm tile. Keep them in one file, `MillAlley_Materials.blend`, one plane per material, named `M01`…`M10` as in the plan.

## Tue Nov 3 · Procedural masks: edge wear and leak streaks

**Goal:** build reusable mask node groups that put wear on edges and streaks under ledges, the custom storytelling layer the plan uses instead of hand-made materials.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Procedural Nodes For Beginners (Blender Tutorial) — Ryan King Art · 24:51-35:53 Texture nodes, 35:53-45:58 Input nodes, 45:58-52:10 Converter (Color Ramp), 1:04:40-end Group nodes (~35 min; 2021 but node behaviour unchanged)
- **Manual:** Bevel Node — Cycles-only edge mask source; also Ambient Occlusion Node
- **Manual:** Color Ramp Node — Also Noise Texture, Gradient Texture, Mix Color, Node Groups

### Steps

1. **Use Cycles.** Render Properties > Render Engine: *Cycles*. The Bevel and Ambient Occlusion nodes only work properly in Cycles. View in Rendered mode (Z > Rendered).
2. **The mask toolkit.** These nodes do most of the work: *Color Ramp* (turns a soft greyscale into a crisp mask), *Math* (Multiply, Subtract, Power), *Noise Texture* (organic break-up), *Gradient Texture* and *Mix Color* (blends two colours by a mask).
3. **Edge mask.** Add *Input* > *Bevel* (Radius 0.02) and *Input* > *Geometry*. Plug Bevel's Normal and Geometry's Normal into a *Vector Math* node set to *Dot Product*. Flat areas give 1, rounded edges less than 1. Add *Math* > *Subtract* with 1 on top (1 − value) so edges become white, then *Math* > *Multiply* by about 10 and tick *Clamp*.
4. **Keep wear on exposed edges.** Add *Input* > *Ambient Occlusion*, take its AO output, and multiply it with the edge mask. Edges tucked into corners now get less wear.
5. **Break it up.** Add a *Noise Texture* (Scale 8, Detail 6) into a *Color Ramp* with the stops pushed close together, and multiply that with the edge mask. Wear now comes in chips instead of a clean line.
6. **Leak streaks.** Add *Texture Coordinate* (use its *Object* output) into *Separate XYZ*, and take Z into a *Color Ramp* so the mask is white at the top of the object and fades to black lower down. Add a second Noise Texture, fed by a *Mapping* node with Scale X 20, Y 20, Z 1, which stretches the noise into vertical drips. Multiply gradient × drips and sharpen with a Color Ramp.
7. **Use the masks.** A *Mix Color* node blends two colours (clean brick and dark grime) using the mask as Factor. Feed the same mask into a Math node to raise roughness where it's dirty.
8. **Make node groups.** Select the edge-mask nodes and press Ctrl+G. Tab in and out of the group. In the sidebar (N > Group) add inputs such as *Radius* and *Amount*, and rename the group `NG_EdgeWear`. Do the same for `NG_LeakStreaks`. Right-click a group > *Mark as Asset* to reuse it in other files.

### Common mistakes

- A mask that's all white or all black. Preview each node (Ctrl+Shift+click) to find the step that fails, and check the Color Ramp stops.
- Seeing nothing from the Bevel or AO node. You're in EEVEE or Material Preview; switch to Cycles Rendered.
- Leak noise stretched sideways. Mapping Scale must be large on X and Y and small on Z.

### Drill: edge wear on a bevelled cube (30 minutes)

Add a cube with a Bevel modifier (Amount 0.03, 3 segments). Give it a dark painted-metal material and use `NG_EdgeWear` to mix in a bright bare-metal colour and lower roughness on the edges.

**Pass when:** wear shows only on outside edges, it breaks into chips rather than a solid line, the inner corners stay clean, and changing the group's Amount input widens or narrows the wear.

### Use it on Mill Alley

Today you build the edge-wear and leak mask groups and drive a test wall module with them. Put the leak streaks under a sill on your B02 drill wall from Week 2 and the edge wear on its corners.

## Fri Nov 6 · Baking procedural materials to tiling textures

**Goal:** bake a node material to image textures UE5 can use, and make sure they tile.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to Texture Bake Procedural Materials in Blender (Tutorial) — Ryan King Art · 0:00-14:58 (skip the paid add-on section), 19:06 total; 2023 video, bake panel unchanged in 5.x
- **Manual:** Render Baking — Bake types, Selected to Active, Margin, output target

### Steps

1. **Why bake.** UE5 can't read Blender nodes. Baking renders the material's result into images (colour, roughness, normal, masks).
2. **Set up Cycles.** Render Properties > Render Engine *Cycles*. Open the *Bake* panel lower down.
3. **Make the target image.** In the material, add an *Image Texture* node, click *New*, name it (`T_M01_Brick_BaseColor`), set 2048 × 2048, untick Alpha. Leave it unconnected and click it last, so it is the *active* node: Cycles bakes into the active Image Texture node.
4. **Bake colour.** Bake Type *Diffuse*, and under Influence tick only *Color* (untick Direct and Indirect) so lighting isn't baked in. Click *Bake*.
5. **Bake anything else through Emission.** For roughness or a mask, temporarily plug that value into an *Emission* shader connected to the Material Output, set Bake Type to *Emit*, and bake into a new image whose Color Space is *Non-Color*. Reconnect the Principled BSDF afterwards.
6. **Bake normals.** Bake Type *Normal*, Space *Tangent*, into a Non-Color image. This captures normal detail from the material's Normal input.
7. **Save the images.** Baked images live only in memory. In the Image Editor, Image > Save As (PNG, 8-bit for colour and masks, 16-bit for normal and height). Unsaved bakes are lost when you close Blender.
8. **Make procedural noise tile.** Image-based layers (the ambientCG brick) already tile. Procedural noise doesn't. To make it seamless, build a `NG_SeamlessNoise` group: UV coordinates > Separate XYZ; multiply X and Y by 6.2832; take Cosine and Sine of each; Combine XYZ from (cos X, sin X, cos Y); feed that into a Noise Texture set to *4D*, with Sine Y into *W*. The noise now wraps perfectly at the UV edges.
9. **Check the tile.** Apply the baked images to a plane, set the Mapping Scale to 3 and look for seams or obvious repeats.

### Common mistakes

- “No active image found”. Click the target Image Texture node before baking.
- Lighting baked into the colour map. Untick Direct and Indirect when baking Diffuse.
- A visible line through the middle of the 3 × 3 check. A procedural layer isn’t seamless; route it through `NG_SeamlessNoise`.

### Drill: bake one brick variant and check it tiles (30 minutes)

Take your M01 brick, add soot with seamless noise and the edge-wear group, and bake Base Color, Roughness and Normal at 2K.

**Pass when:** the three PNGs are saved, a 3 × 3 tile of them shows no seams, and the colour map has no baked shadows.

### Use it on Mill Alley

Today you build brick sets M01 to M04 (soot, limewash peel, infill variants) and bake them to 2K tiling sets. Use the naming `T_M01_Brick_BaseColor`, `T_M01_Brick_Normal`, `T_M01_Brick_ORM` (packed tomorrow).

## Sat Nov 7 · Seamless checks and channel packing

**Goal:** pack several greyscale maps into one texture’s red, green and blue channels, the way game studios save memory.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** DI - Channel Packing with Blender (ARM, ORM Texture Maps) — Digital Impostor · Whole video, 8:41 (2020; Separate/Combine RGB nodes are now named Separate/Combine Color)
- **Manual:** Combine Color Node — Pack three greyscale masks into R, G, B
- **Manual:** Render Baking — Bake each packed mask out to one image

### Steps

1. **Why pack.** Roughness, ambient occlusion and metallic are each one greyscale channel. One RGB texture holds three of them for the cost of one. UE5’s usual order is **ORM**: R = Ambient Occlusion, G = Roughness, B = Metallic.
2. **Build a packing material.** On a plane, add three Image Texture nodes with your AO, roughness and metallic images, each set to *Non-Color*. Add a *Combine Color* node (mode RGB) and plug them into Red, Green and Blue. Plug Combine Color into an *Emission* shader and that into the Material Output.
3. **Bake the pack.** Add a new 2048 image named `T_M01_Brick_ORM`, set its Color Space to *Non-Color*, make it the active node, and bake with Bake Type *Emit*. Save it as PNG.
4. **Pack masks the same way.** For a mask texture: R = edge wear, G = dirt, B = leaks. You’ll use exactly this in Week 10 for prop variations.
5. **Check the pack.** Open the saved file in the Image Editor and use the channel dropdown in its header to view R, G and B separately. Each should look like the map you put in.
6. **Import correctly in UE5.** For packed and mask textures, untick *sRGB* and set Compression to *Masks*. Leave sRGB on only for colour maps.

### Common mistakes

- Packed maps that look washed out. The target image or a source node was sRGB. Everything in a pack is Non-Color.
- Channels in the wrong order. Write the order into the file name or your notes (ORM).

- Forgetting sRGB off in UE5. The roughness will be wrong in engine even though the file is right.

### Drill: pack edge wear, dirt and leaks into one RGB texture (30 minutes)

Bake the three masks from Tuesday's groups to separate images on a test module, then pack them into `T_Test_Masks_RGB`.

**Pass when:** viewing R, G and B separately shows edge wear, dirt and leaks respectively, and the file is 2K, Non-Color and saved.

### Use it on Mill Alley

Today you finish M05 to M10 (setts, mud, concrete, moss, timber, rusted metal), bake each to 2K with an ORM pack, and bake the edge-wear and leak masks for the kit test pieces.

## Sun Nov 8 · Colour attributes (vertex colours) for UE5 blending

**Goal:** paint red, green and blue vertex colours on a mesh and use them in UE5 to blend moss, grime and wet onto walls.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Import Vertex Color from Blender into Unreal Engine 5 — Drawcall · Whole video (Blender 4 + UE5)
- **Manual:** Vertex Paint — Painting colour attributes
- **Manual:** Object Data (Color Attributes) — Color Attributes panel: add, set active/render
- **Manual:** FBX (Legacy) — Exporter page. The manual does not document the colour option; in the export dialog look under Geometry for the vertex colour setting (None / sRGB / Linear) and use sRGB, the default
- **Unreal docs:** Interchange Import Reference in Unreal Engine (UE 5.8 docs) — Common Meshes > Vertex Color Import Option: Replace / Ignore / Override
- **Unreal docs:** FBX Static Mesh Pipeline in Unreal Engine (UE 5.8 docs) — Vertex colours section

### Steps

1. **What it is.** Every vertex can store a colour. UE5 materials read it as three masks: in this project R = moss, G = grime, B = wet. Painting them is how one tiling wall gets unique moss and grime.
2. **Vertices matter.** Colour blends between vertices, so a single big quad gives a smooth blur. Add loop cuts (or a Subdivide) where you want detail. You'll plan this properly in Week 10.
3. **Add a colour attribute.** Object Data Properties > Color Attributes > +. Name it `Col`, Domain *Vertex*, Data Type *Color*.
4. **Start black.** Switch to Vertex Paint mode (the mode menu, top left). Set the brush colour to pure black and use Paint > Set Vertex Colors (Shift+K) to fill the whole mesh black: no moss, no grime, no wet.
5. **Paint one channel at a time.** Set the brush colour to pure red (1, 0, 0) and the brush *Blend* mode to **Add**. Add raises only the red channel, leaving green and blue untouched. Use *Subtract* with the same red to erase. Repeat with green and blue for the other masks.
6. **See it.** In Solid mode, open the shading dropdown and set Color to *Attribute* to see the paint at any time.
7. **Export it.** In the FBX exporter's Geometry section, keep the *Vertex Colors* option at its default (sRGB), not *None*. Pure red, green and blue paint arrives exactly; check one soft gradient in UE5 the first time to see how blends transfer.
8. **Import it in UE5.** In the import options, set *Vertex Color Import Option* to *Replace*. In a test material, add a *Vertex Color* node and use its R output as the Alpha of a *Lerp* between your brick and moss textures.
9. **Painting in UE5 too.** UE5's Mesh Paint mode paints the same channels on placed meshes. You'll mostly paint there; Blender painting is for assets that need baked-in variation.

## Common mistakes

- Painting with Mix blend and pure red. Mix overwrites green and blue to 0, erasing your other masks. Use Add and Subtract.
- Painting a mesh with too few vertices. You get blurry triangles. Add loop cuts first.
- Vertex colours missing in UE5. Check the FBX colour option and UE5's *Replace* import option.

## Drill: paint a moss channel and see it in UE5 (30 minutes)

Make a 4 × 4 m wall with about 8 loop cuts each way. Fill it black, paint red moss along the bottom and under a ledge, export it, and import it into UE5 with a Lerp test material.

**Pass when:** moss appears in UE5 exactly where you painted it, green and blue stay at zero (check in Blender with Solid > Attribute), and the brick still shows where you didn't paint.

## Use it on Mill Alley

Today you build `M_Master_Kit` in UE5 with the vertex height blend and Review 3. Use your drill wall as the test mesh. For the standard check, render your ten materials on cylinders side by side, as Courtyard and Tsushima do, and save the image for the material sheet.

# Week 4 · Nov 9 to 15 · Trim sheets and the first kit family

This week's goal is to make trim sheets, unwrap UVs onto them at the project's texel density, and build Family A as clean modules that snap together.

## Mon Nov 9 · Trim sheet layout and high-poly strips

**Goal:** plan a trim sheet in pixels and model the high-poly strips that Marmoset will bake onto it.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Trim Sheets: A Complete Tutorial — Josh - Blender Bros (Josh Gambrell) · 49:25 total; watch roughly the first 20 min on planning and modeling strips
- **Manual:** Subdivision Surface Modifier — High-poly strips
- **Manual:** Bevel — Support bevels on strip edges

### Steps

1. **What a trim sheet is.** One texture made of horizontal strips (a sill profile, a lintel, a coping stone…). Any mesh can map its long faces onto a strip, so dozens of pieces share one texture. Strips repeat left to right, so a 2 m strip can cover a 6 m string course.
2. **Do the pixel maths.** At 10.24 px/cm a 2048 px sheet covers 200 cm both ways. A strip's height in pixels = its real height in cm × 10.24. Round each strip to a multiple of 8 px and leave an 8 px gap between strips for mip-mapping.
3. **Write the T01 stone plan.** For example: sill front 15 cm → 160 px; sill top 20 cm → 208 px; lintel face 30 cm → 312 px; string course 20 cm → 208 px; coping 30 cm → 312 px; kerb 15 cm → 160 px. Add the gaps and keep the total under 2048. Put the leftover space into a plain tiling stone strip.
4. **Block the strips.** Make one 2 m wide plane per strip at its exact height, stacked in your plan's order: this is the layout. Then make a single 2 × 2 m plane, the *low-poly* target the bake goes onto.
5. **Model each high-poly strip.** On each strip, model the real profile: a drip groove under the sill, a chamfer on the coping, chips and joints. Keep the strip's front face lined up with the target plane and give it some depth behind.
6. **Smooth edges.** Add a *Subdivision Surface* modifier (Ctrl+1, Ctrl+2…) and Shade Smooth. Add support edge loops (Ctrl+R close to an edge) where an edge should stay crisp, or use a Bevel modifier (3 segments) under the Subdivision.
7. **Make strips tile.** Any detail crossing the left or right edge must continue on the other side. Duplicate the finished strip 2 m to the left and right during the bake (Array, Count 3, centred) so the edges get baked from the neighbouring copy.
8. **Name for baking.** Name the high meshes `T01_high` and the target plane `T01_low` (Marmoset pairs them by these suffixes). Export each as FBX.

### Common mistakes

- Strip heights that don't match the real sizes. Then the sill on the sheet won't fit the sill on the mesh without stretching.
- Detail that stops at the strip edge. Use the 3-copy trick in step 7.
- Strips touching each other. Bleeding between strips shows up as thin lines at distance. Keep the 8 px gaps.

### Drill: lay out the T01 stone strips (30 minutes)

Write the T01 plan from step 3 (pixel heights, gaps, order), block every strip as a 2 m plane at its height, and model one strip fully (the sill with its drip groove).

**Pass when:** the strips add up to 2048 px or less, every strip is a multiple of 8 px, and the sill strip is smooth with crisp edges and a groove that reads from 3 m away.

### Use it on Mill Alley

Today you model the T01 stone and T02 metal trim strips and bake them to 2048 sheets in Marmoset. Plan T02 the same way (window frame sections, pipe, rail, bracket, flat metal).

## Tue Nov 10 · UV unwrapping, trims and texel density

**Goal:** unwrap meshes cleanly and place their UVs on a trim strip at exactly 10.24 px/cm.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** UV Unwrapping for Beginners - Blender Tutorial — Ryan King Art · 4:38-26:10 core unwrapping and seams, 36:28-40:47 Follow Active Quads, 45:35-46:22 apply scale (~28 min); July 2026
- **Read:** Texel Density Checker (extension) — IvanVostrikov · Install from Get Extensions; v2026.1.1, Blender 4.2+
- **Manual:** Seams — Mark Seam
- **Manual:** UV Operators — Unwrap, Smart UV Project, Follow Active Quads

### Steps

1. **The UV Editing workspace.** Left: the UV Editor. Right: the 3D viewport in Edit Mode. Selecting faces in 3D shows their UVs on the left.
2. **A checker texture.** In the UV Editor, Image > New, Generated Type *UV Grid*. Put it on a material. Squares on the mesh should be square and the same size everywhere; that's how you spot stretching.
3. **Seams.** Seams are where the mesh is cut so it can lie flat, like the seams of a cardboard box. In edge mode, select edges and use U > *Mark Seam* (red lines). Hide seams in corners and on edges the camera won't see.
4. **Unwrap.** Select all faces (A) and press U > *Unwrap*. Islands appear in the UV Editor.
5. **Follow Active Quads.** For long strips (beams, mouldings), select one face and make it a perfect rectangle in UV space first (select its UV edges and scale with S X 0 or S Y 0 to straighten). Then select the whole strip with that face active and use U > *Follow Active Quads*. The whole strip unrolls as straight rectangles.
6. **Straighten by hand.** Select a row of UV vertices and press S Y 0 (or S X 0) to line them up perfectly.
7. **Texel Density Checker.** Install it from Edit > Preferences > Get Extensions. In the UV Editor sidebar (N), open its tab, set Texture size 2048 and the target density to 10.24 px/cm (1024 px/m; check which unit the panel uses), then *Set TD* with the islands selected. Use *Calculate TD* to check.
8. **Snap onto a strip.** With the trim sheet loaded in the UV Editor, move the island (G Y) so its height sits exactly inside the strip band. Let it run left or right past the edge: the strip tiles. Don't scale it vertically, or the density breaks.
9. **Pack only for unique textures.** U > *Pack Islands* is for meshes with their own texture (heroes, props), not for trims.

### Common mistakes

- Using Smart UV Project on modules. It scatters islands at random sizes. Mark seams and unwrap on purpose.

- Scaling islands to “fill the square”. On a trim, the strip height decides the scale; on a tiling material, the texel density does.
- Flipped islands. Mirrored text and lighting in UE5 come from mirrored UVs. Check with the UV Grid’s letters.

### Drill: unwrap a beam onto a strip at 10.24 px/cm (30 minutes)

Model a 2 m long, 30 cm deep I-beam (C01). Mark seams, unwrap with Follow Active Quads, set 10.24 px/cm with Texel Density Checker, and place its faces on a T02 strip.

**Pass when:** the UV Grid shows square cells with no stretching, Calculate TD reports 10.24 px/cm, and every island sits inside its strip band with straight edges.

### Use it on Mill Alley

Today you make the T03 timber trim sheet and `M_Master_Trim`, and set up the texel density checker. Write your Texel Density Checker settings into the metrics sheet so every module uses the same numbers.

## Fri Nov 13 · Modular walls that snap

**Goal:** build wall modules with exact edges, correct pivots and light-catching bevels, so two copies meet with no gap and continuous brick.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Essential Tips for Modular Building — Grant Abbitt (Gabbitt) · Whole video, 3:32
- **Video:** Modular Walls - Blender to Unreal 5 — Chrismartinartist · Whole video, 14:45: grid-sized walls and pivots into UE5
- **Manual:** Weighted Normal Modifier — Face-weighted normals on bevelled modules
- **Manual:** Smooth By Angle Modifier — Replaces old Auto Smooth; also Object Origin for pivots

### Steps

1. **Edges on the grid.** Every outer edge of a module lands on a 25 cm grid line. Use typed values and Absolute Grid Snap. Check with N > Item: dimensions like 4.0, 0.3 and 4.0, never 3.997.
2. **Pivot bottom-left-back.** Set the origin to the module’s bottom-left-back corner (Shift+S > Cursor to Selected on that vertex, then Set Origin > Origin to 3D Cursor) and move the object to the world origin with Alt+G.
3. **Bevels that catch light.** Add a Bevel modifier, Amount 0.01–0.02, Segments 2, Limit Method *Angle*. Only the outside corners get rounded.
4. **Weighted normals.** Add a *Weighted Normal* modifier below the Bevel (Mode *Face Area*, tick *Keep Sharp*). It makes big flat faces shade flat while the small bevels shade smoothly. This is the “mid-poly” look that needs no normal-map bake.
5. **Smooth by Angle.** Object > *Shade Auto Smooth* adds a *Smooth by Angle* modifier (30°). Keep it at the bottom of the stack.
6. **World-scale UVs.** Unwrap the front face so 2 m in the world = 1 UV unit (the 2 m tile). Put the bottom-left corner of the face at UV 0, 0. Since modules are 100, 200 or 400 cm wide, every module ends on a whole or half tile, and the brick coursing continues into the next module.
7. **Check from both sides.** Two copies placed side by side: no gap, no overlap, no shading seam, and brick that lines up course for course.

### Common mistakes

- A tiny gap or overlap between modules. A vertex is off the grid; check its global position in the Item panel.
- Dark smudges on flat faces. Missing Weighted Normal, or Smooth by Angle placed before the Bevel.

- Brick jumping half a course at the join. The UVs don't start at 0, or the density isn't exactly 10.24 px/cm.

### Drill: two A01 walls that snap (30 minutes)

Model `SM_MillAlley_A_WallFlat_400x400` (400 × 400 cm, 30 cm deep) with bevels, weighted normals, world-scale UVs and a bottom-left-back pivot. Duplicate it, move the copy G X 4 Enter, and apply your brick material.

**Pass when:** the two walls meet with no visible seam in Material Preview, brick courses continue across the join, and the same test passes in UE5 with grid snap 25.

### Use it on Mill Alley

Today you model A01, A02, A06 and A09 with bevels and UVs; the snap test on the 25 cm grid is the "done when". Use this drill's settings as the template for all four.

## Sat Nov 14 · Arches, mouldings and clean topology

**Goal:** model curved openings with clean, game-ready topology: no n-gons, no flipped faces.

**Time:** about 75 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** How to Use the Spin Tool in Blender — Ryan King Art · Rewatch for arches and mouldings (Jan 2026)
- **Manual:** Spin — Arch profiles
- **Manual:** Viewport Overlays — Face Orientation (Geometry section) to catch flipped normals

### Steps

1. **Arch curve with Spin.** In Edit Mode, add a single vertex at the arch's springing point (the left end of the curve). Put the 3D cursor at the arch's centre. In front view, use the Spin tool: Angle 180°, Steps 16. You get a clean half-circle of edges.
2. **Connect it to the wall.** Select the arch edges and the matching wall edges and use Edge > *Bridge Edge Loops* (both loops must have the same number of edges), or F to fill smaller gaps. *Grid Fill* fills a hole with a clean quad grid.
3. **Circular arrays.** For repeated arch bricks or ribs, model one piece, add an Empty at the arch centre, and give the piece an Array with *Relative Offset* off and *Object Offset* set to the Empty. Rotate the Empty by 180 ÷ (count − 1) degrees on Y.
4. **Cleaning booleans.** After applying a boolean, fix its long triangles: J connects two selected vertices with an edge, K is the Knife for drawing new edges, and X > *Dissolve Edges* removes extra ones without leaving holes.
5. **Find n-gons.** Select > Select All by Trait > *Faces by Sides*, Number of Vertices 4, Type *Greater Than*. Everything selected has more than four sides. Cut them into quads or triangles.
6. **Face Orientation.** Overlays > *Face Orientation*. Blue faces point outward, red faces are backwards. Fix red ones with Mesh > Normals > *Recalculate Outside* (Shift+N).
7. **Mouldings along a path.** For a string course, model its cross-section once and extrude it the module's length; give its front faces trim UVs.

### Common mistakes

- Bridge Edge Loops failing. The loops need the same number of edges; add or remove cuts until they match.
- A wobbly arch. The Spin centre wasn't exactly at the arch centre. Snap the cursor to a vertex first.
- Red faces in Face Orientation. Backwards faces render black or invisible in UE5.

### Drill: a clean A03 arch (40 minutes)

Model A03 (300 × 600 cm): wall, round-headed recess 240 cm wide, 25 cm deep, with brick infill at the back. No booleans this time: build the curve with Spin and connect it with Bridge Edge Loops or Grid Fill.

**Pass when:** *Faces by Sides > 4* selects nothing, Face Orientation is all blue, the curve is a clean half-circle, and the module snaps to A01 with no gap.

### Use it on Mill Alley

Today you model A03, A05 door surround, A07 string course and A08 parapet and get them into the engine with materials. A05 is the same method as your drill, with the 240 × 320 cm loading door opening.

## Sun Nov 15 · The Asset Browser and batch export

**Goal:** turn modules into drag-and-drop assets and export a whole family with correct names in one click.

**Time:** about 45 minutes for the lesson and 25 for the drill.

### Watch and read first

- **Video:** Build Scenes FASTER: Mastering the Asset Browser in Blender 5.1 — Grant Abbitt (Gabbitt) · Whole video, 9:03 (Blender 5.1)
- **Manual:** Asset Browser — Mark as Asset, catalogs, drag in
- **Manual:** Asset Libraries: Introduction — Where assets live; also Collections Exporters for batch export

### Steps

1. **One collection per module.** Put each module in its own collection named exactly like the mesh (`SM_MillAlley_A_WallFlat_400x400`), inside a family collection (`Kit_A`).
2. **Mark as asset.** Right-click the object in the Outliner > *Mark as Asset*. Blender makes a preview thumbnail; right-click > *Generate Preview* to refresh it.
3. **Asset Browser.** Change any editor to *Asset Browser* (editor type menu, top left of the area). Pick *Current File* to see this file's assets. Make catalogs (folders) on the left, such as Kit A, Kit B, Props.
4. **Use them from any file.** Edit > Preferences > File Paths > Asset Libraries: add the folder where your kit `.blend` files live. Now every file can drag modules in from that library.
5. **Drag-build.** Drag a module into the viewport. Snapping works while dragging. Build a test wall from several modules.
6. **Batch export.** Give each module collection a Collection Exporter (Collection Properties > Exporters > Add Exporter > FBX legacy) using your UE5 settings, with the file path set to `//Export/` + the module name. File > Export All Collections writes every module in one go.
7. **Import the set into UE5.** Select all the FBXs and import them together into a `Kit_A` folder.

### Common mistakes

- Assets that don't appear in other files. The file must be saved inside a folder registered in Asset Libraries.
- Exported names that don't match. UE5 takes the name from the object or file; keep collection, object and file names identical.
- Exporting the whole scene into one FBX. That gives one giant mesh in UE5. One file per module.

### Drill: drag-build a wall from the Asset Browser (25 minutes)

Mark your A01, A02, A03 and A09 as assets in a `Kit A` catalog, open a new file, and build a 12 m wall section by dragging them from the library.

**Pass when:** the wall builds with no gaps using only drag and snap, and Export All Collections writes one correctly named FBX per module.

## Use it on Mill Alley

Today you model the A04 window, replace the left greybox with the kit and hold Review 4. For the standard check, lay Family A on a neutral backdrop with each piece’s dimensions written beside it, as Le Chlanko does.

# Week 5 · Nov 16 to 22 · Kit families B, C and D

This week's goal is to finish the kit: corners that wrap brick cleanly, window variants from one base, stairs on the step grid, pipes and rails from curves, and ground tiles within budget.

## Mon Nov 16 · Corners, seams and UVs that wrap

**Goal:** make corner pieces where the brick coursing wraps around the corner without a jump.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** UV Unwrapping for Beginners - Blender Tutorial — Ryan King Art · Review 10:22-14:18 (texture repeat, seamless textures) and 40:47-44:09 (projection types, Project From View)
- **Manual:** Mirror Modifier — Mirroring modules
- **Manual:** UV Operators — Unwrap/projection options for matching islands across a corner

### Steps

1. **Why corners are hard.** Two wall faces meet at 90°. If each face is unwrapped separately, their brick courses start at different heights and the corner shows a step.
2. **Model the corner.** B08 is 50 × 380 cm: an L-shape 50 cm along each wall, 30 cm thick, with its pivot at bottom-left-back like every module. Bevel the outside corner 1–2 cm.
3. **Unwrap both faces as one strip.** Put the seam on an edge the camera won't see (the back), not on the corner itself. Unwrap, then straighten. The two faces now sit side by side in UV space, so the courses run around the corner.
4. **Line up V with the walls.** The bottom edge of the corner's UVs must sit at the same V value as the bottom of B01's UVs (V 0 at the floor) and use the same density. Then every course meets.
5. **Mirroring.** For symmetric pieces, model half and use a Mirror modifier. Mirroring is fine for brick and stone; avoid it on anything with readable text or obvious direction (stains running up).
6. **Vertical joins between storeys.** The right building's 380 cm storey isn't a whole number of 8.33 cm courses (45.6), so a module stacked on another can show a half-course step. That's why the B07 floor band sits on that line: it hides the join. Plan your bands for exactly this.
7. **Inner corners.** A06 inner corners are the same idea turned inside out. The seam can sit right in the inside corner, where shadow hides it.

### Common mistakes

- Seams on the visible corner. Move them to the back or the top.
- Corners at a different texel density from the walls. Run *Calculate TD* on both.
- Forgetting the bevel on the corner. A bevelled wall meeting a sharp corner looks wrong in raking light.

### Drill: outer corner B08 against B01 with continuous coursing (30 minutes)

Model B08, unwrap it as one strip, and place it against a B01 wall with the painted brick material.

**Pass when:** from any angle the brick courses run unbroken around the corner, the snap test passes, and Calculate TD matches B01.

### Use it on Mill Alley

Today you model B01, B07, B08 and B09; the snap test is the “done when”. Do B07 (floor band) right after the drill, since it hides the storey joins.

## Tue Nov 17 · Variants from one base: linked duplicates, Knife and Separate

**Goal:** make intact, broken and gutted versions of the steel window from one base, sharing one UV layout.

**Time:** about 60 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Blender 5 Complete Beginners Guide FREE COURSE Part 7 the street — Grant Abbitt (Gabbitt) · 4:52-5:53 Linked Duplicates (Alt+D), ~1 min
- **Manual:** Duplicate Linked — Alt+D: shared mesh data
- **Manual:** Knife Topology Tool — K; also Separate

### Steps

1. **Model the base once.** B03 is a steel window insert 140 × 220 cm with a grid of panes about 25 × 35 cm. Model the frame and glazing bars, then the glass as a separate object. UV the frame onto the T02 metal strips and the glass onto its own small layout.
2. **Shift+D vs Alt+D.** Shift+D makes a separate copy you can edit freely; Alt+D makes a linked copy that shares the mesh. Use Shift+D for variants (their shapes will differ) and Alt+D for identical repeats (the same frame in three bays).
3. **Broken variant.** Duplicate with Shift+D. Delete some glass panes (select faces, X > Faces). On others, use the Knife (K): click points across a pane to draw a crack, Enter to confirm. Select one piece of the cracked pane and press P > *Selection* to make it a separate shard you can move.
4. **Bend a bar.** Turn on *Proportional Editing* (O). Select a glazing bar’s vertices and move them: nearby vertices follow with falloff. Scroll while moving to change the falloff size.
5. **Gutted variant.** Duplicate again, delete all the glass and most glazing bars, leaving the frame and a few bent stubs.
6. **UVs carry over.** Deleting, cutting and moving faces keeps their UVs, so all three variants share the same texture layout.
7. **Name the set.** `SM_MillAlley_B_WindowSteel_140x220_Intact`, `_Broken`, `_Gutted`. Same pivot, same size, so they swap in UE5 without moving.

### Common mistakes

- Using Alt+D for variants. Editing one breaks the others. Check with the user count number next to the mesh name in Object Data properties.
- Knife cuts that don’t go through. Press C while cutting to cut through all faces, or make sure each click lands on an edge.
- Variants with different pivots. They’ll jump when you swap them in the level.

### Drill: three B03 variants sharing one UV layout (40 minutes)

Make the intact, broken and gutted windows.

**Pass when:** all three have the same outer size and pivot, show the same texture placement on the frame, and the broken one has at least five separate shards and one bent bar.

### Use it on Mill Alley

Today you model the B02 window bay and the three B03 variants, and the "done when" is swapping variants cleanly. Test the swap in UE5: right-click a placed window > *Replace Selected Actors with* the other variant; nothing should move.

## Fri Nov 20 · Stairs and the light well on the step grid

**Goal:** build stairs to exact riser and tread sizes with an Array, and fit them into the light well.

**Time:** about 45 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Modifiers for Beginners (Blender Tutorial) — Ryan King Art · Rewatch 24:55-28:42 Array
- **Manual:** Array Modifier — Constant Offset for exact riser and tread steps
- **Manual:** Snapping — Increment snapping to the step grid

### Steps

1. **Know the numbers.** The plan's light well: 120 cm wide, about 200 cm drop, 11 risers of 18 cm and treads of 28 cm. That's a 198 cm drop and 308 cm of run, leaving a landing in the 380 cm length.
2. **One step.** Make a box 120 × 28 × 18 cm with its origin at the top-front edge. Add a small bevel to the nosing (the front edge).
3. **Array the flight.** Add an Array: *Relative Offset* off, *Constant Offset* on with Y 0.28 and Z −0.18, Count 11. That's the whole flight, each step exactly one tread back and one riser down.
4. **Fill underneath.** Give the step some height below so the flight is solid, or model a sloped soffit. Hidden faces can be deleted to save triangles.
5. **Snap to the step grid.** Temporarily set the grid scale to the riser height, or use typed values, to place the landing exactly at −1.98 m.
6. **Walls and landing.** Build the light well walls (concrete) and landing as separate modules with the same pivot rule. Leave room for the C05 handrail posts every 150 cm.
7. **Check with a figure.** Drop the 180 cm figure on the stairs. If the steps look like a ladder or a ramp, the numbers are wrong.

### Common mistakes

- Using Relative Offset for stairs. It moves by the whole step size, including any bevel, and the numbers drift. Use Constant Offset.
- Forgetting the drop doesn't divide evenly. 11 × 18 = 198 cm, not 200. Use 198 everywhere.
- Steps floating off the landing. Check the last step's bottom with N > Item.

### Drill: an 11-step stair that matches the plan (30 minutes)

Build the B06 flight and landing inside a 120 × 380 cm light well.

**Pass when:** there are 11 risers of exactly 18 cm and treads of exactly 28 cm (Edge Length overlay), the bottom tread meets the landing at −1.98 m, and the figure walks it convincingly.

### Use it on Mill Alley

Today you model B04 door bay, B05 ledge and B06 light well, and the right wall becomes fully kit-built. The light well is part of hero H3, so give its edges good bevels: it's a foreground close-up later.

## Sat Nov 21 · Curves for pipes, gutters and rails

**Goal:** draw pipes and rails as curves, give them thickness, bend arrays along them, and convert them to game meshes.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to Use Curves for Beginners (Blender Tutorial) — Ryan King Art · Whole video (June 2025): Bezier curves, bevel depth, converting to mesh
- **Manual:** Curve Geometry — Bevel Depth/Resolution for pipes
- **Manual:** Curve Modifier — Bend arrays along a curve; Convert to mesh

### Steps

1. **Add a Bezier curve.** Shift+A > Curve > Bezier. In Edit Mode each point has two handles. E extrudes a new point. V changes the handle type: *Vector* gives straight segments and sharp corners, *Aligned* or *Auto* give smooth bends.
2. **Give it thickness.** Object Data Properties > Geometry > Bevel: *Round*, Depth 0.05 (a 10 cm drainpipe), Resolution 4. Tick *Fill Caps* to close the ends.
3. **Keep it light.** Under *Shape*, lower *Resolution Preview U* so curves don't become thousands of triangles. Check the count with the Statistics overlay.
4. **Draw the drainpipe.** In front view, place points straight up the wall with Vector handles, then an offset (two 45° bends) where it steps around the ledge, then a shoe at the bottom that kicks out about 45°.
5. **Curve modifier.** For things made of repeated parts (a rail with posts, a chain of brackets), model one part, add an Array to repeat it along X, then a *Curve* modifier pointing at the curve. The array bends to follow the curve.
6. **Convert to mesh.** Object > Convert > *Mesh* when the shape is final. Then unwrap the pipe with Follow Active Quads and place it on the T02 pipe strip.
7. **Snap the pieces.** Make standard lengths (100, 200, 400 cm straights, one elbow, one offset, one shoe) with pivots at their ends so they chain together in UE5.

### Common mistakes

- A pipe that's flat or twisted. Bevel *Depth* is the radius, not the diameter; twisting comes from tilt (Ctrl+T in Edit Mode).
- Huge triangle counts. Lower the curve's resolution before converting.
- Converting too early. Keep the curve until you like the shape; you can't get it back after converting.

### Drill: a drainpipe with an elbow and a shoe (30 minutes)

Draw a 10 cm drainpipe running 4 m up a wall with one offset and a shoe, convert it, and UV it onto a strip.

**Pass when:** it has no twists, uses under about 600 triangles, the bends are smooth, and the UV Grid shows no stretching along its length.

### Use it on Mill Alley

Today you model Family C: beams, drainpipes, the hopper head, brackets, the handrail and gutters. "Done when" is pipes running full height with no gaps, so give every pipe piece end pivots and test the chain in UE5.

## Sun Nov 22 · Ground tiles and triangle budgets

**Goal:** model a ground tile with gentle unevenness that still tiles, and keep it within its triangle budget.

**Time:** about 45 minutes for the lesson and 30 for the drill.

## Watch and read first

- **Video:** How to Use the Displacement Modifier (Blender Tutorial) — Ryan King Art · Whole video, about 7-8 min (2022)
- **Manual:** Displace Modifier — Light displacement on subdivided planes; Subdivide
- **Manual:** Viewport Overlays — Statistics overlay (Guides) for triangle counts

## Steps

1. **Start with a grid.** Shift+A > Mesh > Grid, and in the Adjust Last Operation panel set X and Y subdivisions (try 20 × 20) and Size 4 m for D01.
2. **Statistics overlay.** Overlays > *Statistics*. It shows vertices, faces and triangles for the scene and the selection. D01's budget in the plan is up to 4,000 triangles.
3. **Protect the edges.** Tiles must meet flat at their borders. In Edit Mode, select every vertex except the outer border, and in Object Data > Vertex Groups click + and *Assign* to make a group called `Inner`.
4. **Gentle displacement.** Add a *Displace* modifier, set its Vertex Group to `Inner`, click *New* texture and pick a soft noise (Clouds). Keep Strength small (0.03–0.05 m) so it's a subtle dip, not hills.
5. **By hand.** For specific dips (where puddles will sit, or worn wheel tracks), use Proportional Editing (O) and move a few vertices down.
6. **Spend triangles where they show.** Apply the modifier, then use Decimate (*Planar* mode) to merge flat areas, or dissolve edge loops away from the dips. Keep detail where the outline changes.
7. **Check the tiling.** Place four copies in a 2 × 2 grid. The borders must line up exactly in height.

## Common mistakes

- Bumpy tile edges. Without the vertex group, borders move and copies don't meet.
- Displacement that's too strong. Real setts under mud are nearly flat; the material and decals do the rest.
- Ignoring the budget. Ground covers the whole alley; a heavy tile multiplies fast.

## Drill: D01 under 4,000 triangles (30 minutes)

Make D01 (400 × 400 cm) with protected flat borders, a soft dip for a puddle and gentle unevenness elsewhere.

**Pass when:** Statistics shows under 4,000 triangles, four copies meet with no steps at the borders, and the dip reads in Rendered view with a low sun.

## Use it on Mill Alley

Today you model Family D, replace the remaining greybox and hold Review 5 with a kit-only screenshot. For the standard check, lay out every finished piece on a modular-elements sheet, as Cottage and Lemon do.

# Week 6 · Nov 23 to 29 · Sculpting the skybridge

This week's goal is the full high-to-low workflow on the hero piece: a clean base mesh, a sculpted high poly, a game-ready low poly with UVs, a clean bake, and a clay render that shows off the sculpt.

## Mon Nov 23 · Base meshes for sculpting

**Goal:** build base meshes that sculpt well and can double as the low poly, saving a retopology step.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to Use the MULTIRESOLUTION Modifier in Blender — The CG Essentials · 7:59 in total. Watch 0:40 (Subsurf vs Multires) through 3:38 (Sculpt Base Mesh) and 6:40 (Delete Higher)
- **Manual:** Multiresolution Modifier — Subdivide vs Simple, Sculpt/Render levels, Apply Base. This is the high-poly side of the plank
- **Manual:** Subdivision Surface Modifier — Why clean, evenly spaced quads subdivide predictably. Read the levels and topology notes before you build the base mesh

### Steps

1. **Plan high vs low first.** The *low poly* is what goes into UE5 (the skybridge's budget is under 40,000 triangles). The *high poly* holds the sculpted detail, baked into normal and AO maps. Decide which parts get sculpted (cladding boards, window frames) and which are mid-poly with bevels (girders from C01).
2. **Clean quads, even size.** Sculpting subdivides every face. Faces of very different sizes give uneven detail. Aim for roughly square quads of similar size, with no triangles or n-gons.
3. **Model the base.** For a board (15 × 2.5 × 250 cm): a box with loop cuts along its length every 10–15 cm and one or two across its width. Apply scale (Ctrl+A > Scale).
4. **The reuse trick.** Keep a copy of this base mesh. Because Multires keeps the original as level 0, the base you sculpt on can become the low poly later. For planks this is all the retopology you need.
5. **Test subdivision.** Add a Subdivision Surface modifier and view levels 1 to 3. If corners balloon into blobs, add support loops near the ends.
6. **Lay out the whole skybridge.** Frame, cladding boards (an Array from Week 2, now real separate boards), window frames and the girder deck, all matching the greybox volume (500 cm span, 300 cm tall tier, 40 cm deck).

### Common mistakes

- Long thin faces. They sculpt as stretched streaks. Add loop cuts until faces are close to square.
- Unapplied scale. Sculpt brushes behave differently along stretched axes.
- Sculpting everything. Only what's seen up close needs it; the rest can use the trim sheets.

### Drill: a plank base mesh ready to sculpt (30 minutes)

Model a 250 cm board with even quads, plus a duplicate saved as `Plank_low`.

**Pass when:** all faces are quads of similar size, scale reads 1, 1, 1, and Subdivision level 3 keeps the board's shape with no blobby ends.

### Use it on Mill Alley

Today you plan the skybridge from reference and model its base mesh (frame, cladding boards, windows, girder deck). “Done when” is matching the greybox volume, so check it against your greybox in Camera 1.

## Tue Nov 24 · Sculpt Mode: brushes, Multires and masking for wood damage

**Goal:** sculpt believable rot, splits and splinters into timber boards.

**Time:** about 60 minutes for the lesson and 45 for the drill.

### Watch and read first

- **Video:** Sculpting for Complete Beginners! (Blender Tutorial) — Ryan King Art · Uploaded October 2024. Watch 3:55 to 21:00 (Sculpt Mode, brushes, strength/radius, symmetry, remesh, dyntopo) and skip the fish project
- **Video:** Sculpting damage on stone in Blender — Artisans of Vaul · A short demo of breaking edges with Scrape and Clay Strips. Use the same moves for rot and splinters on the board
- **Manual:** Mesh Sculpt Brush Assets — The Essentials brush library. Look up Draw, Clay Strips, Crease Polish and Crease Sharp, Scrape/Fill, Trim, Scrape Multiplane, and Mask
- **Manual:** Trim Gesture Tools — Box Trim, Lasso Trim, Line Trim and Polyline Trim. Difference mode makes clean broken plank ends
- **Manual:** Mask — Mask menu operators (invert, grow, clear) so you can protect areas while you scrape splinters

### Steps

1. **Enter Sculpt Mode.** Use the *Sculpting* workspace or the mode menu. Brushes are assets in the *Essentials* library: pick them from the asset shelf at the bottom of the viewport. The trim tools (Box, Lasso, Line and Polyline Trim) are in the toolbar on the left.
2. **Brush basics.** F changes the radius; Shift+F changes the strength; hold Ctrl to invert a brush (dig instead of build). Turn off X symmetry in the header (on by default): wood damage isn’t symmetrical.
3. **Cut broken ends first.** The *Box Trim* and *Lasso Trim* tools cut the mesh itself. Use them on the base mesh to snap off board ends at an angle, before adding Multires.
4. **Add Multires.** Add Modifier > Generate > *Multiresolution*, and click *Subdivide* three or four times. Sculpt at the highest level; level 0 stays your clean base.
5. **The wood toolkit.** *Scrape/Fill* flattens and wears surfaces; *Crease Sharp* cuts sharp grooves for grain and cracks (*Crease Polish* for softer ones); *Clay Strips* builds up warped or swollen areas; *Draw* adds soft bumps; *Snake Hook* pulls out splinters at broken ends.
6. **Masking.** Paint a mask on areas you want to protect (they turn dark), so brushes only affect the rest. Ctrl+I inverts the mask; Alt+M clears it.
7. **Work big to small.** Start with large warps and sagging, then splits and broken ends, then grain cracks along the length. Stop when it reads from Camera 1’s distance; fine detail that never shows wastes time.
8. **Check under real light.** In Solid mode, try a MatCap (shading dropdown > Lighting > MatCap) with a raking angle. Damage that looks deep head-on can vanish from the side.

### Common mistakes

- Symmetry left on. Every crack appears twice, mirrored.
- Uniform noise everywhere. Rot is concentrated at ends, joints and where water sits; leave quiet areas.
- Sculpting on level 0. Detail must go on the highest Multires level or it changes the low poly.

### Drill: sculpt one rotten board (45 minutes)

Use your plank base mesh: snap one end off with a trim tool, add Multires, and sculpt warping, grain cracks, a rotten end and two splinters.

**Pass when:** the board reads as old timber from 3 m away in a MatCap, the damage is uneven (heavier at the broken end), and Multires level 0 is unchanged.

### Use it on Mill Alley

Today you sculpt the skybridge in Blender: warped and broken boards, rot, sagging and splintered window frames. Sculpt a few "master" boards in detail and reuse them with rotation and flips rather than sculpting every board.

## Fri Nov 27 · Retopology, game meshes and two UV sets

**Goal:** get the sculpt down to a game-ready mesh and unwrap it into texture sets.

**Time:** about 75 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Retopology for Beginners (Blender Tutorial) — Ryan King Art · Uploaded June 2024. Shrinkwrap and face-snapping retopo, start to finish
- **Manual:** Decimate Modifier — Collapse ratio and Planar mode. The fastest route to a plank under 300 tris
- **Manual:** Shrinkwrap Modifier — Project a low-poly cage onto the sculpt
- **Manual:** Snapping — Snap to Face (Project Individual Elements) for manual retopo
- **Manual:** UV Operators — The Unwrap and Pack Islands sections (margin, rotation) for the two 4K UV sets

### Steps

1. **Pick a method per part.** Planks and frames: reuse Multires level 0 (you planned for this on Monday). Rocks and rubble: *Decimate*. Odd organic shapes: manual retopology.
2. **Reuse the base.** Duplicate the sculpted board. On the copy, set the Multires viewport level to 0 and apply it (or delete the modifier). That copy is the low poly. If a trimmed end changed the silhouette, add a few cuts so the low poly follows it.
3. **Decimate.** Add a *Decimate* modifier (Collapse) and lower the Ratio until the silhouette starts to break, then go back up a little. Apply it.
4. **Manual retopology.** Hide the high poly from selection. Make a new mesh, turn snapping to *Face Project*, and add a *Shrinkwrap* modifier targeting the high poly. Place vertices over the surface and fill faces. The *Retopology* overlay (Overlays > Retopology) shows your new faces over the high poly.
5. **Watch the silhouette.** Triangles go where the outline changes. Flat areas need very few.
6. **Two UV sets.** The plan gives the skybridge two 4K sets: *cladding* and *structure*. Give each a different material slot (Material Properties > +, assign faces), then unwrap each group.
7. **Pack Islands.** In the UV Editor, select a set's islands and press Ctrl+P (*Pack Islands*). Set a margin of about 0.005–0.01 so islands don't bleed into each other, and turn on rotation for tighter packing.
8. **Even density.** Run Texel Density Checker across each set. Visible faces should have the same density; hidden undersides can be smaller.

### Common mistakes

- Low poly silhouettes that don't match the high. The bake can't fix silhouette; add geometry there.
- Islands touching. Packing margin too small; you get colour bleed in UE5's lower mip levels.
- Islands scaled at random. One plank looks sharp, the next blurry.

### Drill: a sculpted plank under 300 triangles (30 minutes)

Make a low poly of yesterday's board by reusing level 0 plus a few cuts at the broken end, and unwrap and pack it.

**Pass when:** Statistics shows under 300 triangles, its outline follows the sculpt's broken end, and the UV Grid is undistorted with islands packed and not touching.

### Use it on Mill Alley

Today you retopo the skybridge low poly and UV it into two 4K sets. The “done when” is under 40,000 triangles with clean UVs; keep a running count with the Statistics overlay.

## Sat Nov 28 · Preparing meshes for baking in Marmoset

**Goal:** export high and low meshes so Marmoset bakes them cleanly the first time.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Baking Props in Marmoset Toolbag 5 — Art of Pilgrim · 16:10, uploaded December 2024. 0:37 Quick Loader, 3:00 Geometry settings, 8:00 naming, 11:02 fixing skew
- **Manual:** Triangulate Modifier — Triangulate the low poly before export so Blender and Marmoset agree on the tangent basis
- **Manual:** Object Origin — Set Origin options for matching pivots between the _high and _low objects
- **Marmoset docs:** Baking Attributes (Quick Loader) — Quick Loader section: name pairs Name_high and Name_low (not case-sensitive). Everything before the suffix becomes the bake group, and anything after it is a variation
- **Marmoset docs:** Baking Tips & Tricks — Triangulate before import, split UVs at hard edges, and keep the low poly close in size to the high
- **Marmoset docs:** The Toolbag Baking, Texturing, and Rendering Handbook — By Steph Everett. Covers Quick Loader’s _low/_high requirement, skew and cage

### Steps

1. **Matching names.** Name each pair `<Part>_high` and `<Part>_low` (e.g. `Plank01_high`, `Plank01_low`). Marmoset groups meshes into bake groups by these suffixes, so each high bakes only onto its own low.
2. **Same place, same pivot.** High and low must overlap exactly in space. Don’t move one without the other.
3. **Triangulate the low.** Add a *Triangulate* modifier to the low poly and export with modifiers applied. The bake and UE5 then use the same triangles, so normal maps shade correctly.
4. **Smoothing.** Keep the low poly smooth-shaded with hard edges only at UV seams. A hard edge without a UV seam causes a visible bake line.
5. **No overlapping UVs.** Every island in a set needs its own space (unless you deliberately offset mirrored copies by a whole UV tile).
6. **Export.** Two FBX files: all highs in one (`Skybridge_high.fbx`), all lows in another (`Skybridge_low.fbx`). Apply modifiers. The high export can be heavy; that’s fine.
7. **Bake in Marmoset.** Create a new bake project, drop both files into the baker (the Quick Loader slots), set the output size to 4096 and pick maps: Normal, AO, Curvature, Thickness, Height, ID. Set the normal map to suit Unreal (DirectX / flip Y). Adjust the cage offset until red error areas in the preview disappear, then bake.

### Common mistakes

- Black splotches or waviness. The cage is too tight or too loose; adjust the offset and paint it per area.
- Details baking onto the wrong plank. Pieces near each other need separate bake groups (correct _high/_low names).
- Seams in the normal map. Hard edges without a UV seam, or an untriangulated low.

### Drill: bake the test plank with no artefacts (30 minutes)

Export the plank’s high and low with correct names and triangulation, and bake Normal and AO at 2K in Marmoset.

**Pass when:** the baked low poly looks like the sculpt in Marmoset's viewport, with no black splotches, no seams across the plank, and no detail from elsewhere.

### Use it on Mill Alley

Today you bake the skybridge in Marmoset and start its texture project (peeling red and green paint, rot, rust on the girders). Everything from this drill applies, just with more bake groups.

## Sun Nov 29 · Clay renders

**Goal:** render a clean clay image of a sculpt, the standard way to show sculpting skill without textures.

**Time:** about 40 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Easy Three Point Lighting Setup in Blender! — Ghost Studios · Uploaded June 2025. 1:30 infinite background, 3:23 three-point lights, 7:25 camera
- **Manual:** Viewport Shading — Solid mode lighting: MatCap vs Studio, and single-colour grey for a quick clay look
- **Manual:** Light Objects — Area light power and size for key, fill and rim

### Steps

1. **A backdrop.** Add a plane, extrude its back edge upward (E Z 3), and bevel the joint with several segments. That's a seamless studio sweep.
2. **Clay material.** One material for everything: Principled BSDF, Base Color mid-grey (about 0.5), Roughness about 0.6. Give the backdrop a slightly darker grey.
3. **Three-point lights.** Add three Area lights: a *key* light 45° to the side and above (strongest), a *fill* light on the other side at about a third of the key's strength, and a *rim* light behind the object to outline its silhouette.
4. **Camera.** A longer lens (50–85 mm) avoids distortion. Frame the object with some space around it.
5. **Render.** Cycles for quality (sculpt detail reads through shadow), 128–256 samples with denoising on. Render at the size your portfolio uses (for example 1920 × 1080).
6. **The quick version.** For progress shots, Solid mode with a MatCap and Viewport Render Image (View > Viewport Render Image) is good enough.

### Common mistakes

- A pure-white clay. Highlights burn out and hide the detail. Mid-grey shows sculpting best.
- Flat front lighting. The key light must rake across the surface to show damage.
- A cluttered background. Keep it plain; the sculpt is the subject.

### Drill: a clay render of the skybridge sculpt (30 minutes)

Set up the backdrop, clay material and three lights, and render the skybridge high poly.

**Pass when:** the damage reads clearly, the silhouette is outlined by the rim light, nothing is pure white or pure black, and the background is plain.

### Use it on Mill Alley

Today you finish texturing the skybridge, import it, place it and hold Review 6. For the standard check, add this clay render to a clay sculpt sheet, as Cottage and Courtyard do.

# Week 7 · Nov 30 to Dec 6 · Hero assets and catch-up

This week's goal is to sculpt masonry, damage metal convincingly and break glass, then use the two buffer days to close any gaps from Weeks 1 to 6.

## Mon Nov 30 · Sculpting masonry and chipped edges

**Goal:** sculpt bricks and stone with believable chips and wear, for the chimney's corbelled cap.

**Time:** about 60 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Sculpting damage on stone in Blender — Artisans of Vaul · Chipped stone edges with Scrape and Clay Strips
- **Video:** Complete Beginner's Guide to Sculpting in Blender 4.4 — Grant Abbitt (Gabbitt) · Uploaded April 2025, on the brush-asset version of Sculpt Mode. Optional refresher on brush setup
- **Manual:** Mesh Sculpt Brush Assets — Clay Strips for brick faces. Trim and Scrape/Fill for flat chipped planes
- **Manual:** Remesh (Sculpt Mode) — Voxel Remesh (Ctrl-R) to even out density before you chip the edges
- **Manual:** Trim Gesture Tools — Lasso Trim in Difference mode for big chunks knocked off the corbel cap

### Steps

1. **Start with a block.** Model one brick at real size (about 23 × 11 × 7.5 cm) with a small bevel. Masonry sculpts best from a simple block with even density.
2. **Voxel Remesh.** In Sculpt Mode, the *Remesh* panel in the header (or R to set the voxel size, then Ctrl+R) rebuilds the mesh as an even grid of small quads. Use a voxel size around 2–3 mm for a brick. Every face is now the same size, which brushes need.
3. **Chip the edges.** Use *Lasso Trim* to slice off a corner (it cuts a clean flat break), then *Scrape/Fill* lightly over the break to soften it. Hold Ctrl with it to dig small pits.
4. **Wear the faces.** *Clay Strips* (inverted with Ctrl) carves shallow dips; *Draw* with a low strength adds small pocks. Old brick is worn most on its exposed arrises (edges) and faces, not in the mortar bed.
5. **Remesh again when needed.** Big trims leave uneven faces; remesh to tidy them, at the cost of softening fine detail. Do big shapes, remesh, then fine detail last.
6. **Make variants fast.** Duplicate a finished brick and change one or two chips. Five bricks are enough to build a corbel course by rotating and flipping them.
7. **Corbel cap.** Model the cap as stepped rings of blocks (each course a little wider than the one below), then sculpt chips along the course edges where weather hits hardest. Bake it onto a simple low poly like the skybridge.

### Common mistakes

- Chips that look like bites. Real chips are flat breaks; use trims, not round brushes.
- Every edge chipped equally. Wear is uneven; leave some edges sharp.
- Remeshing at the end. It wipes out fine detail; remesh before the last pass.

### Drill: five chipped bricks (40 minutes)

Sculpt five different bricks from one base: two lightly worn, two with corner chips, one badly broken.

**Pass when:** each reads as a different brick from 1 m away in a MatCap, chips have flat break faces, and at least one edge on each brick stays crisp.

### Use it on Mill Alley

Today you model the chimney and red end building and sculpt the corbelled cap. Use your five bricks as the stamp set for the cap's courses; that also gives you P04 loose bricks for Week 8.

## Tue Dec 1 · Hard-surface damage: dents and torn metal

**Goal:** dent and tear sheet metal for the rusted loading doors.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** How to Use the Displacement Modifier (Blender Tutorial) — Ryan King Art · 0:56 enough geometry, 2:22 add the modifier, 3:18 procedural texture (use Noise for dents)
- **Manual:** Displace Modifier — Texture, Strength, Midlevel and vertex-group limits for local dents
- **Manual:** Knife Topology Tool — Cutting tear lines into the sheet metal
- **Manual:** Split — Split or Rip along the knife cut so the torn flap can bend away

### Steps

1. **A panel with enough geometry.** Make a plane at the door panel's size and subdivide it (Edit Mode > right-click > Subdivide, or a Grid primitive) so it has vertices to dent. Add Solidify (Thickness 0.003, 3 mm sheet steel).
2. **Overall waviness with Displace.** Add a *Displace* modifier above Solidify. Click *New* to make a texture; in the Texture properties set it to *Clouds* with a large size. Keep Strength small (0.005–0.01). Sheet metal is never perfectly flat.
3. **Local dents.** For a dent where something hit the door, select a vertex, turn on Proportional Editing (O) and move it inward, scrolling to set the dent's size. A sharp dent: small falloff. A broad bow: large falloff.
4. **Torn edges with the Knife.** Press K and click a jagged path along a corner (zigzags of uneven length). Enter to confirm. Select the corner piece and delete it, or separate it (P) to bend it outward with proportional editing.
5. **Bend the flap.** Select the torn flap's vertices, place the 3D cursor on the bend line, set the pivot to *3D Cursor* (the pivot menu in the header, or the period key pie) and rotate it outward.
6. **Supporting details.** Rivets (small cylinders arrayed along the edge), hinges and a frame from T02 trims make damage believable by contrast.
7. **Bake or keep.** For a unique 2K door (H6), use this as the high poly and bake it onto a simple low poly. Dents with large shapes can stay in the low poly geometry.

### Common mistakes

- Displace with no effect or spikes. Too few vertices for subtle waves, or Strength too high.
- Clean, straight tears. Real torn metal zigzags and curls; vary every segment.
- Dents everywhere. A few big story-telling hits read better than uniform noise.

### Drill: one dented door panel (40 minutes)

Make a 120 × 200 cm steel door panel with overall waviness, two dents, one torn and bent corner, and a row of rivets.

**Pass when:** the panel catches light unevenly in Rendered view with a low light, the tear zigzags, and the bent flap has visible thickness.

### Use it on Mill Alley

Today you texture the chimney cap and model and texture the H6 rusted loading doors (they sit in the 240 × 320 cm A05 opening). The drill is one leaf of those doors.

## Fri Dec 4 · Breaking glass with Cell Fracture

**Goal:** smash window panes into shards with the Cell Fracture extension and place them convincingly.

**Time:** about 40 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Breaking glass using cell fracture - Blender tutorial — Imperfect Render · Uploaded November 2024. Plan the fracture lines with annotations, then set up Cell Fracture. Skip the rigid-body section
- **Read:** Cell Fracture (Blender Extensions platform listing) — Install it via Edit > Preferences > Get Extensions
- **Manual:** Cell Fracture (Blender 4.1 Manual, legacy add-on page) — The last manual page before it became an extension. Location: 3D Viewport > Object > Quick Effects > Cell Fracture
- **Manual:** Separate — By Loose Parts, if shards are joined. Cell Fracture already outputs separate objects

### Steps

1. **Install it.** Edit > Preferences > Get Extensions, search “Cell Fracture”, Install. It’s free.
2. **Make a pane.** A plane at the pane’s size (about 25 × 35 cm for the steel windows), with Solidify applied for 4–6 mm thickness. Cell Fracture needs a closed mesh.
3. **Run it.** With the pane selected, run Cell Fracture from the Object menu’s *Quick Effects*. Set *Source Limit* to about 20 for 20 shards.
4. **Control where it breaks.** Draw annotation strokes (the Annotate tool in the toolbar) where the impact was, and choose the annotation (*Grease Pencil / Annotation*) source. Shards cluster where you drew, like a real impact point.
5. **Check the output.** Each shard becomes its own object. Hide the original pane.
6. **Make the break story.** Delete some shards so there’s a hole; leave jagged ones still in the frame; move a few to the sill and ground below, rotated and lying flat.
7. **Join for export.** Select the shards that stay together (in the frame) and join them (Ctrl+J). Loose ones become P10 scatter props.

### Common mistakes

- “Mesh not manifold” or no shards. Apply Solidify first; the pane must be closed.
- Too many shards. Hundreds of tiny pieces cost triangles and read as dust. Twenty is plenty.
- Shards floating. Use snapping to *Face* when placing shards on the sill.

### Drill: a smashed pane with 20 shards (30 minutes)

Fracture one pane into about 20 shards around an impact point, leave jagged pieces in the frame, and drop five onto a sill.

**Pass when:** there’s a clear impact point, the shards in the frame are joined into one object, five lie flat on the sill with no floating, and the total is under 1,000 triangles.

### Use it on Mill Alley

Today you make the H7 broken glazing variants and `M_Master_Glass`, assemble the H3 light well vignette and hold Review 7. Your drill feeds the broken B03 variant directly.

## Sat Dec 5 · Buffer: redo a failed or skipped drill

**Goal:** close the gaps from Weeks 1 to 6 before the second half starts.

**Time:** as long as you need; no new lesson.

### Steps

1. **List your gaps.** Go through Weeks 1 to 6 in the tracker. Any drill not ticked, or ticked without meeting its *Pass when*, goes on today's list.
2. **Priority order.** These carry into everything later, so redo them first if they're on your list:
   - The 4 × 4 m wall round trip to UE5 (Week 1, Sat): scale and pivots.
   - Two A01 walls that snap (Week 4, Fri): modular edges and UVs.
   - Unwrap a beam onto a strip at 10.24 px/cm (Week 4, Tue): texel density.
   - Bake one brick variant and check it tiles (Week 3, Fri): baking.
   - Bake the test plank with no artefacts (Week 6, Sat): Marmoset baking.
3. **Redo, don't reread.** Do the drill from a blank file, without the old file open, against its pass check.
4. **Write one line per drill** in the tracker's notes: what went wrong last time, and what fixed it.

### Use it on Mill Alley

The Mill Alley plan's buffer today is for anything late from Weeks 3 to 7. Do the kit or hero task first if it's behind; drills second.

## Sun Dec 6 · Buffer: free practice

**Goal:** rest your schedule or build speed; no new lesson.

**Time:** optional.

### Steps

1. **If you're behind,** use today for the Mill Alley plan's catch-up.
2. **If you're on track,** pick one speed drill and time it:
   - Block out a new building facade from a photo in 30 minutes (Week 2 skills).
   - Model and UV a new trim-sheet piece in 20 minutes (Week 4).
   - Sculpt a damaged stone block in 20 minutes (Week 7).
3. **Learn the shortcuts you still look up.** Keep a list of every menu you opened by hand this week; find its shortcut in the Blender Manual and use it all next week.

### Use it on Mill Alley

Everything should be in the scene by now: kit, heroes, glazing. Take a dated progress screenshot of Camera 1 for the process sheet.

# Week 8 · Dec 7 to 13 · Props and debris

This week's goal is to make props fast: variants by randomizing, rubble with physics, cables with Geometry Nodes, cheap debris on shared atlases, and a prop sheet with triangle counts.

## Mon Dec 7 · Fast props: random variants and Simple Deform

**Goal:** make many convincing timber variants quickly, and warp plywood with a modifier.

**Time:** about 45 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to Randomize Transformations in Blender — Nik Kottmann · 5:22 in total. Two worked examples
- **Video:** Using the SIMPLE DEFORM MODIFIER in Blender to Bend, Twist, and Taper Objects! — The CG Essentials · Recorded in 2020, but the modifier is unchanged. Watch the Bend part
- **Manual:** Randomize Transform — Object > Transform > Randomize Transform: location, rotation and scale ranges with a seed
- **Manual:** Simple Deform Modifier — Bend and Twist modes, axis and limits for warped plywood
- **Manual:** Array Modifier — Relative offset for board runs before you randomize

### Steps

1. **One good base.** Model a plank at the plan's size (15 × 5 × 240 cm) with a small bevel and loop cuts every 20 cm along its length, so it can bend. UV it onto the T03 timber strip.
2. **Duplicate and vary.** Shift+D five copies. Give each a different change: cut an end at an angle (Knife or Bisect), snap one in two, chip a corner, shorten one.
3. **Randomize Transform.** Select several copies, Object > Transform > *Randomize Transform*. Small values only: Location 0.02, Rotation 3°, Scale 0.05. It breaks the "copy-paste" look when they lie in a pile.
4. **Simple Deform.** Add Modifier > Deform > *Simple Deform*. *Bend* curves the mesh, *Twist* twists it, *Taper* narrows one end. Set the axis and a small angle (5–15°). It bends around the object's origin, so move the origin to change where the bend happens.
5. **Warp the plywood.** P01 is 122 × 244 × 1.8 cm. Subdivide it along both directions, add Simple Deform Bend on one axis and a second Simple Deform on the other at a smaller angle. Plywood bows and curls at the corners when wet.
6. **Delamination.** Duplicate a corner region of the top face (Shift+D, P to separate) and lift it slightly with proportional editing: the top ply peeling away.
7. **Apply when happy.** Apply the modifiers before export so the shape is fixed.

### Common mistakes

- Simple Deform does nothing or kinks. The mesh needs loop cuts along the bend direction.
- Randomizing scale too much. Planks of wildly different thickness look wrong; real timber is a standard size.
- Bending around the wrong point. Move the origin, or set an empty as the *Origin* in the modifier.

### Drill: six plank variants in 30 minutes

From one base plank, make six variants: two straight, two warped (Simple Deform), one snapped, one with an angled end.

**Pass when:** done within 30 minutes, all six share the same UVs and material, and in a pile none looks like an obvious copy.

### Use it on Mill Alley

Today you make P01 plywood, P02 planks and P03 offcuts and place them in a prop gallery level. P03 offcuts are just your snapped variants cut shorter.

## Tue Dec 8 · Rubble: fracturing, physics piles and shared atlases

**Goal:** make natural-looking rubble clusters that share one texture atlas.

**Time:** about 60 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Using the CELL FRACTURE Add-On for Blender - Step by Step Tutorial! — The CG Essentials · 0:25 enable, 0:51 fracture, 1:36 noise and variation, 3:12 more fragments. Stop before physics at 4:21
- **Video:** How to UV Unwrap Multiple Objects onto One UV Map! — Blender Inferno · Blender 4.2, short. Several props sharing one atlas
- **Manual:** Cell Fracture (Blender 4.1 Manual, legacy add-on page) — Option reference. In 4.2+ it is an extension under Object > Quick Effects
- **Manual:** Using UV Maps — Several UV maps and shared images, as background for the shared rubble atlas
- **Manual:** UV Operators — Pack Islands across several objects in Edit Mode together

### Steps

1. **Pieces.** Start from your five sculpted bricks (Week 7). Make half bricks and chunks by running Cell Fracture on a brick (Source Limit 3–5), and keep the best pieces.
2. **Drop them with physics.** Select the pieces and use Object > Rigid Body > *Add Active*. Select the ground plane and *Add Passive*. Stack the pieces loosely above the ground and press Space to play: they fall and settle naturally.
3. **Freeze the result.** At a frame you like, select the pieces and use Object > Rigid Body > *Apply Transformation*. Then remove their rigid bodies.
4. **Add fill.** Small chunks and a low mound of mud/mortar dust underneath tie the pile to the ground.
5. **Shared atlas.** Join every rubble piece into one object (Ctrl+J), unwrap, and use *Pack Islands* so they all share one texture. If you need them separate again, P > *By Loose Parts*: each piece keeps its spot on the atlas.
6. **Bake.** Bake the high-poly sculpted pieces onto this atlas in Marmoset like the skybridge, so one 1K texture covers the whole cluster.
7. **Build clusters.** Join the settled pile into a cluster mesh (P05) and save individual bricks too (P04) for hand placement.

### Common mistakes

- Pieces exploding apart. They were overlapping at the start; spread them out first.
- Rubble piled like a cone of identical bricks. Mix sizes: a few big pieces, many small ones, plus dust.
- Each piece with its own texture. That's many draw calls; one atlas for the whole set.

### Drill: one rubble cluster on a shared atlas (40 minutes)

Drop about 15 pieces into a pile with rigid bodies, freeze it, add a mortar-dust base, and pack all pieces onto one atlas.

**Pass when:** the pile looks settled (no floating or interpenetrating pieces), it's under 2,500 triangles, and all pieces use one material and one UV atlas without overlaps.

### Use it on Mill Alley

Today you make P04 bricks and P05 rubble piles (sculpt, bake, atlas); “done when” is three rubble clusters ready. Make the other two clusters by dropping the same pieces again with a different starting arrangement.

## Fri Dec 11 · Cables and hoses with Geometry Nodes

**Goal:** make a cable that sags naturally between two points using a curve and a simple Geometry Nodes setup.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** 3 ways to make cables out of 3D curves in Blender — Jen Abbott Creates · Uploaded February 2025, short. Options at 0:03, 1:06 and 2:40
- **Manual:** Curves — Bezier and NURBS editing for drawing the cable path
- **Manual:** Curve Geometry (Bevel) — Bevel depth and resolution for a quick round cable
- **Manual:** Curve to Mesh Node — Profile curve and Fill Caps for the Geometry Nodes version
- **Manual:** Proportional Editing — Pull the middle points down smoothly to sag the cable by hand

### Steps

1. **Draw the path.** Shift+A > Curve > Bezier. In Edit Mode, put the two end points on the brackets (vertex snapping helps) and add a middle point (select both, right-click > Subdivide).
2. **Sag by hand.** Move the middle point down. Set the handles to *Auto* (V > Auto) so the curve forms a smooth catenary-like dip. Long spans sag more; a tight cable barely dips.
3. **Geometry Nodes.** Open the *Geometry Nodes* workspace, select the curve, click *New*. You see *Group Input* → *Group Output*.
4. **Curve to Mesh.** Add (Shift+A) *Curve to Mesh* between them. Add a *Curve Circle* node into its Profile input; set radius 0.01 (2 cm cable) and resolution 6 to keep it cheap.
5. **Shade smooth.** Add *Set Shade Smooth* after Curve to Mesh.
6. **Edit stays live.** Moving the curve’s points still updates the cable. Change the circle radius for a hose (P22).
7. **Make it a mesh.** Object > Convert > *Mesh* when final, then unwrap with Follow Active Quads onto a tiling cable texture.
8. **In UE5 too.** For long runs, UE5 Spline Mesh actors bend one cable segment along a spline in the editor. Export a straight 100 cm cable segment for that.

### Common mistakes

- A cable that looks like a stiff rod. Add a bit more sag and a slight sideways drift.
- Triangle-heavy cables. Lower the curve resolution (Object Data > Shape) and the circle resolution.
- No UVs after converting. Unwrap after conversion.

### Drill: a sagging cable between two brackets (30 minutes)

Model two simple wall brackets 3 m apart and run a cable between them with Geometry Nodes, sagging about 30 cm.

**Pass when:** the ends sit exactly in the brackets, the sag is smooth, the converted mesh is under 300 triangles, and the UV Grid isn’t stretched.

### Use it on Mill Alley

Today you make P06 pipes, P07 cable spline, P08 fallen gutter and P09 broken frame; “done when” is the cable spline usable in the level. Export the straight segment for UE5’s spline tool too.

## Sat Dec 12 · Small debris: cards, instancing and atlases

**Goal:** make litter and small debris that costs almost nothing, using flat cards, low-poly shapes and one atlas.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** How to Bake Normal Map Decals in Blender (Tutorial) — Ryan King Art · Uploaded April 2025. Watch 8:30 to 12:36 (plane, texture, UVs, bake settings, save). The same bake gives normal-mapped debris cards
- **Manual:** Render Baking — Selected to Active normal bake from a modelled piece onto a flat card
- **Manual:** Collection Instancing — Reuse litter pieces as instances
- **Manual:** UV Operators — Pack Islands for the shared litter atlas

### Steps

1. **Cards.** Paper, leaves and flat litter are single planes with a texture and an alpha (cut-out). A few loop cuts let you crumple them slightly with proportional editing.
2. **Low-poly solids.** Bottles and cans are cylinders with 8 sides; a bag is a crumpled subdivided plane. Detail comes from the normal map, not geometry.
3. **Bake detail onto cards.** Sculpt a crumpled paper high poly (or use Displace + noise), bake its normal onto the flat card in Marmoset, just like the planks.
4. **One atlas for the set.** Put every piece's UVs onto one 1K texture: lay the cards and small props out together and pack them. One material for the whole litter set.
5. **Instancing.** Repeats of the same piece should be linked duplicates (Alt+D) or collection instances. In UE5, the same mesh placed many times is cheap.
6. **Count as you go.** Select a piece and read Statistics: a can at about 60 triangles, a card at 2–8.

### Common mistakes

- Modelling labels and creases in geometry. Paint or bake them; keep the mesh tiny.
- Pieces at different texture densities on the atlas. Some look sharp, others blurry; check with Texel Density Checker.
- Cards visible from behind as invisible. Use a two-sided material in UE5 for paper and leaves.

### Drill: a litter set of eight pieces under 1,000 triangles (40 minutes)

Make eight pieces (for example a can, a bottle, a crushed can, a bag, two paper cards, a cardboard scrap and a cigarette pack) on one 1K atlas.

**Pass when:** all eight together are under 1,000 triangles, they share one material, and they read clearly at 2 m in Material Preview.

### Use it on Mill Alley

Today you make P10 glass shards, P11 dead branches, P12 corrugated sheet, P15 litter and P16 paper; "done when" is all P1 and P2 small props done.

## Sun Dec 13 · Prop sheet presentation

**Goal:** present props like a portfolio prop sheet: clean backdrop, good light, wireframe and triangle counts.

**Time:** about 40 minutes for the lesson and 40 for the drill.

## Watch and read first

- **Video:** Blender: Render Your Wireframes | Tutorial — PIXXO 3D · Uploaded January 2025, Blender 4.3
- **Video:** How to render wireframe on shaded material and make wireframe transparent in Blender — blenderian · 0:29 Wireframe modifier, 1:00 material, 2:46 Eevee/Cycles render, 3:40 transparent wire
- **Manual:** Viewport Overlays — The Wireframe overlay and Statistics, for reading triangle counts
- **Manual:** Wireframe Modifier — Render-time wireframe on top of the shaded prop
- **Manual:** Text — Text objects for tri-count labels in the render

## Steps

1. **Lay out the props.** On your studio sweep from Week 6, arrange the props in neat rows, largest at the back. Leave space for labels.
2. **Light it.** Same three-point setup as the clay render, slightly softer (bigger area lights).
3. **The beauty pass.** Render with full materials in Cycles or EEVEE.
4. **The wireframe pass.** Switch to Solid shading, turn on Overlays > *Wireframe* at full opacity, hide the grid, axes and other overlays, and use View > *Viewport Render Image*. Same camera, same framing as the beauty pass.
5. **Triangle counts.** Select each prop and read its triangle count from the Statistics overlay. Add a label under it: Shift+A > Text, Tab to edit the text ("Plywood · 400 tris"), and rotate it to face the camera. Or add labels in an image editor afterwards.
6. **Assemble.** Put the beauty and wireframe versions side by side or as a before/after slider image.
7. **Credits.** List any third-party material or prop you adapted (Poly Haven, ambientCG, Blendkit…) with its source.

## Common mistakes

- Different camera between passes. The wireframe must line up with the beauty shot exactly.
- Unreadable labels. Use a plain font large enough to read on a phone.
- Busy backgrounds. The props are the subject; keep the backdrop plain.

## Drill: one prop sheet render (40 minutes)

Render a prop sheet of this week's props: beauty pass, matching wireframe pass, and triangle-count labels.

**Pass when:** the two passes line up exactly, every prop has a readable label with its count, and the image reads on a phone screen.

## Use it on Mill Alley

Today you make P17 junction box and P18 lamp bracket (then P3 props if time) and hold Review 8 with a prop sheet: your drill is that sheet. For the standard check, show props with triangle counts and a set-dressing wireframe, as Le Chlanko and Lemon do, and credit any third-party assets you adapted, as Indian street does.

# Week 9 · Dec 14 to 20 · Foliage

This week's goal is game foliage built the AAA way: a leaf atlas, cards made from it, ivy and grass placed with tools, wind-ready plants, and a breakdown sheet that shows all of it.

## Mon Dec 14 · Leaves for an atlas

**Goal:** model leaves and lay them out on a 2048 atlas ready to bake into opacity, normal and colour maps.

**Time:** about 60 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Blender Secrets - Bake Displacement Maps from Camera View — Blender Secrets · Short. Bake detail laid out on a plane through an orthographic camera, the same setup as the leaf atlas
- **Manual:** Cameras — Orthographic type and Orthographic Scale for framing a 2048 atlas exactly
- **Manual:** Render Baking — Normal and other passes baked from leaf geometry to the atlas plane

### Steps

1. **What an atlas is.** One texture holding many leaves, twigs and blades. Every foliage card samples part of it, so all your plants share one material.
2. **Model a leaf.** Add a plane, add a few loop cuts along its length and across, and in top view move the vertices into the leaf's outline (ivy: 3–5 lobes; buddleia: long and narrow; grass: a thin tapered blade). Give it a slight bend down the midrib with proportional editing.
3. **Colour.** Use a CC0 leaf scan, or paint a base colour with vertex colours: darker at the base, lighter toward the tip, a slightly lighter midrib.
4. **Lay out the atlas.** Make a 2 × 2 m plane as the atlas area. Arrange your leaves and small clusters on it without overlapping, with gaps between them. Big leaves and the clusters you'll use most get the most space. Leave a strip along one edge for grass blades.
5. **Orthographic camera (for a quick preview).** Add a camera above the plane, set its type to *Orthographic* and its *Orthographic Scale* to 2 so it frames the atlas exactly; set the render to 2048 × 2048. A render with a transparent background (Render > Film > *Transparent*) previews the opacity map.
6. **Bake properly.** For the real atlas, name the leaves `Atlas_high` and the plane `Atlas_low`, and bake in Marmoset: Normal, Opacity (from the high's coverage), Base Color (from vertex colour or texture) and Thickness (use it for translucency).

### Common mistakes

- Leaves overlapping on the atlas. They bake into one blob.
- Perfectly flat leaves. A slight bend gives the normal map its shading.
- Wasting space. Empty atlas space is wasted texture memory; scale important elements up.

### Drill: ivy and buddleia leaves on one atlas (40 minutes)

Model three ivy leaves, two buddleia leaves and two grass blades, and lay them out on the atlas plane with clusters of 3–5 ivy leaves.

**Pass when:** nothing overlaps, at least two-thirds of the area is used, and the orthographic preview render shows clean leaf shapes on a transparent background.

### Use it on Mill Alley

Today you model the leaves (ivy, buddleia, grass, weeds) and bake the foliage atlas in Marmoset; “done when” is an atlas with opacity, normal and translucency maps.

## Tue Dec 15 · Ivy: Ivy Gen and leaf cards along curves

**Goal:** grow ivy down a wall and drainpipe and turn it into game-ready cards.

**Time:** about 60 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Blender’s Built-In IvyGen Is EASY & FAST | Tutorial — PIXXO 3D · Uploaded January 2023 (pre-extension). The workflow is the same once the extension is installed
- **Read:** Ivy Gen (Blender Extensions platform listing) — Install it in 4.2+ via Preferences > Get Extensions
- **Manual:** Ivy Gen (Blender 4.0 Manual, legacy add-on page) — Sidebar (N) > Create tab > Ivy Generator panel. Snap the 3D cursor to the start vertex, then Add New Ivy or Add Default Ivy, then Update Ivy
- **Manual:** Curves — Draw your own ivy strand as a curve if Ivy Gen does not suit

### Steps

1. **Install Ivy Gen.** Edit > Preferences > Get Extensions, search “Ivy Gen”, Install.
2. **Grow it.** Place the 3D cursor where the ivy starts (Shift+right-click on the parapet), select the wall or pipe it should climb, then open the Sidebar (N) > *Create* tab > *Ivy Generator* panel and click *Add New Ivy*. Use *Update Ivy* after changing settings. Change the settings (size, gravity weight, adhesion, leaf density) and regenerate until the shape cascades like the photo.
3. **Use it as a guide.** Ivy Gen’s leaves are simple quads with their own texture. For the game, keep its *stems* (convert them to mesh, or keep them as curves) and replace the leaves with your atlas cards.
4. **Make leaf cards.** A card is a plane (or two crossed planes) UV’d onto a cluster of 3–5 leaves on your atlas. Make three or four card types (small, medium, big cluster).
5. **Scatter cards along a curve with Geometry Nodes.** Draw a stem curve along the drainpipe. Add a Geometry Nodes modifier: *Resample Curve* (Count about 40) → *Instance on Points* with a card from *Collection Info* (tick *Pick Instance* to use all card types). Add *Random Value* nodes into Rotation and Scale for variety, and *Realize Instances* before the output.
6. **Face the light.** Real ivy leaves face outward from the wall toward the light. Rotate the cards so they point away from the wall rather than at random.
7. **Convert and export.** Apply the modifier (or Object > Convert > Mesh) and export as one mesh per ivy strand.

### Common mistakes

- Ivy that climbs straight up like a ladder. Add randomness and gravity; it hangs and spreads.
- Leaves facing the wall. Cards must face out.
- Using Ivy Gen’s leaves directly. They don’t use your atlas, so they won’t match the rest of the foliage.

### Drill: an ivy strand down a drainpipe (40 minutes)

Grow an ivy strand from a parapet down a drainpipe with Ivy Gen, then replace its leaves with atlas cards scattered along its stem.

**Pass when:** the strand hangs and wraps the pipe naturally, all leaves are atlas cards facing outward, and the strand is under 3,000 triangles.

### Use it on Mill Alley

Today you make F01 ivy cards and vine splines and place H4 ivy down the left wall and drainpipe. “Done when” is ivy breaking the parapet line like the photo, so check it from Camera 1.

## Fri Dec 18 · Geometry Nodes scattering: grass and weed clumps

**Goal:** build a reusable Geometry Nodes setup that generates grass and weed clumps.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** EASY GRASS in Blender with Geometry Nodes! — The CG Essentials · 0:49 blade cards, 5:32 transparency, 12:03 Distribute Points on Faces, 13:50 Instance on Points, 15:36 randomize. Recorded in 2021, but the node names are unchanged
- **Manual:** Introduction (Geometry Nodes) — The modifier, node groups and fields basics
- **Manual:** Distribute Points on Faces Node — Random vs Poisson Disk, density, seed
- **Manual:** Instance on Points Node — Pick Instance from a collection, plus rotation and scale inputs

### Steps

1. **The idea.** Geometry Nodes generates geometry from rules. Here: scatter points on a surface, put a grass card on each point, randomize.
2. **The base.** Add a small Circle or disc (about 30 cm across), fill it, and add a Geometry Nodes modifier (*New*).
3. **Distribute Points on Faces.** Add it after Group Input. Set *Density* (try 200) and change *Seed* for a different clump.
4. **Instance on Points.** Add it, plug the points in, and feed *Instance* from a *Collection Info* node pointing at a collection of grass blade cards. Tick *Separate Children* and *Pick Instance* so it picks randomly from your cards.
5. **Randomize.** A *Random Value* node (type *Float*, Min 0.7, Max 1.3) into Scale. A *Random Value* (type *Vector*) into Rotation, with Z from 0 to 6.28 (a full turn) and small X and Y tilts (±0.2).
6. **Expose controls.** Drag sockets onto the Group Input (Density, Seed, Scale range). They appear on the modifier panel, so you can make variants without opening the nodes.
7. **Realize and export.** Add *Realize Instances* before the Group Output, then apply the modifier to export a clump mesh.
8. **Reuse.** The node group works on any mesh: put it on a ledge strip and you get grass along the ledge.

### Common mistakes

- Grass standing perfectly straight. Add the small X and Y tilt.
- Seeing only one blade type. *Pick Instance* isn’t ticked.
- Thousands of blades. Clumps for games use tens of cards, not hundreds; watch the triangle count.

### Drill: a grass clump generator (30 minutes)

Build the node setup with exposed Density, Seed and Scale controls, and make three different clumps from it by changing only those controls.

**Pass when:** the three clumps look different, each is under 800 triangles, and they all use your atlas material.

### Use it on Mill Alley

Today you make F02 moss clumps, F03 grass and weeds and F05 dead leaves, all paintable with UE5’s foliage tool. Use your generator for F03 and feed it dead-leaf cards for F05.

# Sat Dec 19 · Bushes and saplings, with wind colours

**Goal:** build a small tree and a bush from branches and leaf cards, and paint vertex colours UE5 can use for wind.

**Time:** about 60 minutes for the lesson and 40 for the drill.

## Watch and read first

- **Video:** Create quick branches using Skin Modifier in Blender — ItsPaulTodd · Short. Blender 3.5, but the workflow is unchanged
- **Video:** Stylized foliage with Wind in Unreal Engine 5 & Blender — ALL THE WORKS · Watch 17:28 (layered wind material) and 24:27 (UE5 tips)
- **Manual:** Skin Modifier — Branch structures from edges, plus per-vertex radius (Ctrl-A)
- **Manual:** Vertex Paint — Paint the wind-weight gradient (dark at the trunk, bright at the tips)
- **Unreal docs:** World Position Offset Material Functions — SimpleGrassWind: WindIntensity, WindWeight, WindSpeed, AdditionalWPO. Feed a Vertex Color channel into WindWeight

## Steps

1. **Branches from curves.** Draw the trunk and main branches as Bezier curves with a Bevel depth. In Edit Mode, select points and press Alt+S to shrink the radius toward the tips, so branches taper.
2. **Or the Skin modifier.** Make a skeleton of connected vertices (extrude with E), add a *Skin* modifier and a Subdivision Surface. Select vertices and press Ctrl+A to scale the skin thickness at each point. Good for knotted, organic trunks.
3. **Convert and simplify.** Convert to mesh and lower the resolution; saplings need few sides (5–6) on thin branches.
4. **Add leaf cards.** Place atlas cards at the branch ends, either by hand or with Instance on Points on the branch tips. Buddleia grows from cracks in walls: short, arching stems with leaf clusters at the ends.
5. **Wind mask.** UE5 bends foliage using a mask: 0 at the base (doesn't move), 1 at the tips (moves most). Store it in vertex colour red.
6. **Make the gradient with Geometry Nodes.** Add a Geometry Nodes modifier: *Position* → *Separate XYZ* → Z into a *Map Range* (From Min 0, From Max = the plant's height, To 0–1) → *Store Named Attribute* (type *Color*, domain *Point*, name `Col`). Apply the modifier. Check it in Solid mode > Color > Attribute.
7. **In UE5.** In the foliage material, plug *Vertex Color* R into the *SimpleGrassWind* node's *WindWeight* input, and its output into World Position Offset. The base stays still and tips sway.

## Common mistakes

- The whole plant swaying, roots included. The wind mask must be 0 at the base.
- Branches as thick at the tip as at the base. Taper with Alt+S or Ctrl+A on the skin.
- Too many leaf cards. Gaps between clusters are part of a believable silhouette.

## Drill: one sapling with wind colours (40 minutes)

Build a 2 m sapling (trunk, 4–6 branches, leaf cards) and store a base-to-tip wind gradient in vertex colour red.

**Pass when:** the attribute view shows black at the base fading to red at the tips, the sapling is under 8,000 triangles, and its silhouette has gaps you can see through.

## Use it on Mill Alley

Today you make F04 buddleia, F06 brambles and F07 saplings, all with wind in UE5. Test the wind material on the drill sapling first.

## Sun Dec 20 · The foliage breakdown render

**Goal:** present foliage the way portfolio breakdowns do: atlas, cards and assembled plant side by side.

**Time:** about 30 minutes for the lesson and 45 for the drill.

### Watch and read first

- **Read:** Creating 3D Scene of Decaying Orthodox Church in Swamp — 80 Level (Nicolas Pablos) · Published September 2026. Read the Vegetation section: atlases, high-poly bakes to cards, and assembled plants. Brickwork and Props are relevant too
- **Manual:** Cameras — Orthographic camera for the side-by-side atlas, card and plant sheet
- **Manual:** Viewport Overlays — Wireframe overlay to show card density

### Steps

1. **Panel 1: the atlas.** Show the base colour and opacity (or normal) of the atlas flat, as images.
2. **Panel 2: the cards.** Lay a few cards flat facing the camera in Solid mode with *Texture* colour, Wireframe overlay on, so viewers see how few polygons carry the leaves.
3. **Panel 3: the plant.** A beauty render of the assembled ivy, buddleia or sapling on your studio backdrop, lit from behind a little so leaf translucency glows.
4. **Matching style.** Same backdrop and light for every panel; labels with triangle counts.
5. **Assemble.** Put the three panels in one image, left to right in build order.

### Common mistakes

- Showing only the final plant. The breakdown's value is showing *how* it's built.
- No triangle counts. Foliage efficiency is a big part of what reviewers look for.

### Drill: the foliage sheet (45 minutes)

Make a three-panel sheet for the ivy and one other plant.

**Pass when:** it shows the atlas, the cards with wireframe, and the finished plant, with triangle counts, and reads on a phone screen.

### Use it on Mill Alley

Today you make F08 background trees and the H5 far-opening mass and hold Review 9. For the standard check, this sheet is the foliage breakdown with atlas, as in Courtyard, Lemon and Windmill.

# Week 10 · Dec 21 to 27 · Decals and vertex paint

This week's goal is the storytelling layer: decals for leaks, cracks and stains, meshes dense enough to vertex paint, and RGB masks that give one prop many looks.

## Mon Dec 21 · Decal geometry

**Goal:** make mesh decals: thin planes floating just off a surface, mapped onto a decal atlas.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Create a Decal Atlas — FlippedNormals · 18:03, a free chapter of 'Sci-Fi Game Environment in Blender & UE4'. Building and mapping a decal atlas
- **Manual:** Solidify Modifier — Offset the decal plane off the wall surface (or move it along normals)
- **Manual:** Using UV Maps — Second UV map or atlas UVs for the decal planes
- **Unreal docs:** Using Mesh Decals — Keep the geometry just in front of the surface, feather the edges for soft fades, and use WPO offset to fix intersections

### Steps

1. **Two kinds of decal.** *Projected decals* are UE5 Decal actors that project a texture onto whatever is behind them; you don't need Blender for those. *Mesh decals* are real planes you model so they follow a shape exactly (around an arch, along a sill). This lesson is mesh decals.
2. **The decal atlas.** One 2048 texture holding leaks, grime patches, cracks and stains, each with soft alpha edges. You'll fill it tomorrow; for now, plan the slots like a trim sheet.
3. **Make a decal plane.** Add a plane where the decal goes. Snap it to the wall (snap to *Face*), then move it out 0.5–1 cm (G Y −0.005) so it never flickers against the wall (z-fighting).
4. **Follow the surface.** For curved or uneven surfaces, add loop cuts and a *Shrinkwrap* modifier targeting the wall, with an *Offset* of 0.005. The decal hugs the surface at a fixed distance.
5. **UVs onto the atlas.** Unwrap the plane and place its UVs over the matching element on the atlas (like trims). A leak under a sill: a long vertical plane mapped to a long vertical streak.
6. **Edge fade.** The alpha in the atlas fades each element's edges, so no hard plane edge shows. Keep the plane a bit bigger than the visible stain.
7. **Material in UE5.** Mesh decals use `M_Master_Decal` (Material Domain *Deferred Decal*, DBuffer with colour, normal and roughness), so a leak can darken and wet the brick without hiding its bumps.

### Common mistakes

- Decals flickering. They're too close to the surface; offset at least 0.5 cm.
- Decals floating visibly off curved surfaces. Use Shrinkwrap rather than a flat plane.
- Hard rectangular edges. The texture's alpha must fade to zero before the plane's edge.

### Drill: three decal planes on a wall (30 minutes)

On a test wall with a sill, place a leak decal under the sill, a grime patch at the base and a crack decal across an arch (Shrinkwrap).

**Pass when:** none flicker in Rendered view up close, the arch decal follows the curve, and each plane's UVs sit on a planned atlas slot.

### Use it on Mill Alley

Today you build decals X01 to X04 and `M_Master_Decal`; "done when" is decals projecting cleanly on brick. Use mesh decals for the leaks under sills (X01) and projected decals in UE5 for the soot gradients (X02).

## Tue Dec 22 · Baking decal maps from sculpted detail

**Goal:** sculpt cracks and leaks and bake them into decal textures, including Blender's own Selected to Active bake.

**Time:** about 60 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** How to Bake Normal Map Decals in Blender (Tutorial) — Ryan King Art · 1:26 model the decal, 8:30 to 12:36 bake to a plane, 13:40 apply
- **Video:** Blender Secrets - Bake Displacement Maps from Camera View — Blender Secrets · Height map from a sculpted plane with an orthographic camera
- **Manual:** Render Baking — Normal bake and Emit-based height bake from the sculpted plane to the 1K atlas
- **Manual:** Multiresolution Modifier — Sculpt the cracks on a subdivided plane
- **Unreal docs:** Decal Materials — Deferred Decal domain, DBuffer blend modes (Translucent, Alpha Composite)

### Steps

1. **Sculpt on a plane.** A subdivided plane (or Multires) sized like the decal slot. Use *Crease Sharp* to cut cracks, *Scrape/Fill* for spalled brick faces, and a soft *Draw* (inverted) for water-worn channels.
2. **A flat low.** A plain plane over the sculpt, UV'd to the decal's slot on the atlas.
3. **Bake in Blender: Selected to Active.** In Cycles, select the high (sculpt) first, then Shift-click the low so it's *active*. In the Bake panel, tick *Selected to Active* and set *Extrusion* (or *Max Ray Distance*) a little more than the sculpt's depth. Bake Type *Normal* into a new 1K Non-Color image on the low's material.
4. **Height too.** For height, bake the sculpt's position into an Emit bake (or bake height in Marmoset, which does it directly). Height drives parallax or the decal's normal strength.
5. **Opacity.** Paint the decal's mask: white where the crack or stain is, fading to black. Use the Image Editor's paint mode, or bake a mask from a Color Ramp on the sculpt's height.
6. **Save and pack.** Save each map, then pack them into the atlas slot (colour, normal, opacity/roughness).

### Common mistakes

- Black areas in the normal bake. The Extrusion is too small for the sculpt's depth.
- The wrong object active. The low (target) must be the active, last-selected object.
- Hard edges where the decal's alpha ends. Fade the mask well inside the slot.

### Drill: a crack decal baked to 1K (30 minutes)

Sculpt a crack running across three bricks on a plane, bake its normal to a 1K image with Selected to Active, and paint its opacity mask.

**Pass when:** the normal map shows the crack's depth with no black patches, the mask fades softly, and a decal plane using them reads as a real crack on a brick wall.

### Use it on Mill Alley

Today you build decals X05 to X08; "done when" is the full decal library in the engine. X04 brick damage and cracks use exactly this drill's method.

## Fri Dec 25 · Buffer (Christmas)

**Goal:** no lesson today.

### Steps

1. Take the day off. If you want something light, re-watch any resource from Weeks 8 to 10 you skimmed.

### Use it on Mill Alley

The Mill Alley plan is also on buffer today.

## Sat Dec 26 · Meshes made for vertex painting

**Goal:** give wall modules vertices where moss, grime and wet need to change, without wasting triangles.

**Time:** about 40 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Import Vertex Color from Blender into Unreal Engine 5 — Drawcall · Uploaded January 2025, Blender 4. The full round trip plus a preview material
- **Video:** Blender Vertex Painting Tutorial | Master Color Attributes in Blender — Jouleconcepts · Uploaded September 2025. Colour attributes and game-asset tips
- **Manual:** Loop Cut — Add edge loops where moss and grime gradients need vertices
- **Manual:** Vertex Paint — Painting colour attributes
- **Manual:** FBX (Legacy) — The FBX exporter page in 5.2. Check the vertex colour option under Geometry so colour attributes reach UE5
- **Unreal docs:** FBX Import Options Reference — Vertex Color Import Option: Replace takes the FBX colours, Ignore keeps colours painted in the editor
- **Unreal docs:** Activating and Using Mesh Paint Mode — Painting extra grime per instance in UE. Nanite meshes do not show per-instance vertex colour

### Steps

1. **How painting works in UE5.** Mesh Paint mode stores colours per placed copy, so every A01 in the alley can be painted differently. But colour only changes at vertices: on a single 4 × 4 m quad, a stroke becomes a huge blurry gradient. Per-copy paint doesn't show on Nanite meshes, so keep the wall modules you paint in UE5 non-Nanite (or paint them in Blender before export).
2. **Plan where paint goes.** Grime rises from the ground (needs horizontal loops near the base), moss sits on ledges and in corners, and wet streaks fall from sills (needs vertical loops under openings).
3. **Add loops on purpose.** Use Ctrl+R with several cuts: about every 25 cm in the bottom metre of a wall, every 50–100 cm higher up. Put vertical loops at window edges, so leaks can start there.
4. **Keep the grid even.** Even quads blend smoothly. Long thin triangles give jagged blends.
5. **Check the budget.** Painting density is worth it on hero walls near Camera 1. Background walls can stay light.
6. **Check export.** Export one module with its colour attribute (even if it's black) with *Vertex Colors* on in the FBX settings and confirm UE5 imports it with *Replace*; then test painting it in Mesh Paint mode.
7. **Paint in UE5.** In Mesh Paint mode, paint *Vertex Color* with R, G or B enabled one at a time (moss, grime, wet), matching your `M_Master_Kit`.

### Common mistakes

- A beautiful blend in Blender that looks blocky in UE5. Same cause: not enough vertices where the blend changes.
- Loops everywhere. Triangles double with each cut; add them only where paint goes.

- Painting all three channels at once. Turn channels on one at a time.

### Drill: a facade with enough vertices for smooth moss and grime (40 minutes)

Add paint loops to your A01 and B02 modules, import them, and paint grime rising 1 m from the ground and a wet streak under a sill in UE5.

**Pass when:** the grime fades smoothly with no stair-steps, the wet streak starts right under the sill, and each module stays within its triangle budget (A modules up to 3,000, B up to 4,000).

### Use it on Mill Alley

Today you vertex paint both facades: moss where water sits, grime rising from the ground, wet under leaks. "Done when" is no visible tiling in Camera 1. Fix any module that paints blocky by adding loops and reimporting (Week 2's round trip).

## Sun Dec 27 · RGB masks: one prop, many looks

**Goal:** use one packed mask texture to make several versions of a prop from a single mesh and texture set.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Adding dirt to models in Unreal Engine using Masks — FastTrack Tutorials · Variation masks and how a material uses them
- **Manual:** Render Baking — Bake the edge-wear, dirt and chip masks, then pack them into R, G and B
- **Manual:** Using UV Maps — The mask shares the barrel's main UVs
- **Unreal docs:** Setting Up a Texture Blended Material for Vertex Weights Painting — Use the 3-way R/G/B Lerp graph with your packed mask texture instead of vertex colour

### Steps

1. **The idea.** A mask texture's channels hold R = edge wear, G = dirt, B = paint chips (packed as in Week 3). The material blends extra colours and roughness using those masks; changing a few numbers gives a new look.
2. **Bake the masks.** For the barrel (P13), bake edge wear and dirt from your Week 3 node groups (or from Marmoset curvature and AO), and paint the paint-chip mask. Pack them into `T_P13_Barrel_Masks`.
3. **Set up in Blender.** In the material, separate the mask's channels (*Separate Color*) and use each as the Factor of a *Mix Color*: base paint colour → mix with bare metal using R → mix with dirt using G → mix with a second paint layer using B.
4. **Make controls.** Group the setup (Ctrl+G) and expose inputs: paint colour, dirt amount, wear amount.
5. **Three versions.** Duplicate the material three times and change only the inputs: blue barrel lightly worn, red barrel very dirty, rusted barrel with wear turned up.
6. **The same in UE5.** Build it once as a master material with parameters, then make *Material Instances* for each look. Changing an instance's values costs nothing at runtime.

### Common mistakes

- Masks in sRGB. They must be Non-Color in Blender and sRGB off in UE5.
- All three looks too similar. Change colour *and* wear and dirt amounts.
- Baking a new texture per variant. The point is one texture set, many looks.

### Drill: three variants of one barrel from one mask (40 minutes)

Make the barrel's mask texture and three materials driven by it (clean painted, dirty, rusted).

**Pass when:** all three use the same mesh, the same base textures and the same mask, and they look like three different barrels in a side-by-side render.

## Use it on Mill Alley

Today you place decals (leaks under sills, rust bleed, soot, peeling edges) and hold Review 10. For the standard check, render the three barrels side by side: the RGB-mask variation on a reusable prop, as Le Chlanko and Lemon do.

# Week 11 · Dec 28 to Jan 3 · Set dressing and composition

This week's goal is to dress the scene like an environment artist: debris clusters built from your props, compositions that lead the eye, and weeds scattered where they would really grow.

## Mon Dec 28 · Kitbashing debris clusters

**Goal:** build debris piles from props you already have, and decide when to join pieces and when to instance them.

**Time:** about 40 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Quick Tip: Quick Set Dressing for Unreal Engine using Blender — Learn Squared (Joe Garth) · 4:56, watch all. Building dressing clusters in Blender for UE (2020, workflow unchanged)
- **Video:** Blender Tutorial - Full Guide to Instancing Collections! — Blender Made Easy · Optional. Collection instances vs real geometry (2021; the instancing UI is the same in 5.x)
- **Manual:** Join — Ctrl+J merges the debris pieces into one mesh for a single exported cluster (one draw call, one pivot)
- **Manual:** Collection (Instancing) — The non-destructive option: keep the cluster as a collection and place instances; pair with Duplicate Linked (Alt+D) for shared mesh data

### Steps

1. **Link your props in.** File > *Link* (or the Asset Browser) brings props from their own files without copying them. Edit the source file and the change shows everywhere.
2. **Build from big to small.** Start with one or two large pieces (a plank, a broken frame), then medium (bricks, a pipe), then small (litter, shards, leaves) tucked into the gaps. Real debris collects against walls and in corners.
3. **Ground it.** Pieces should sink slightly into the ground and each other. Nothing floats. Use snapping to *Face* with *Align Rotation to Target* to lay pieces on surfaces.
4. **Join or instance?** *Instance* (Alt+D or collection instances) when the same piece repeats many times or you want to keep editing it. *Join* (Ctrl+J) into one cluster mesh when it's a fixed composition you'll place as a unit in UE5. Joined clusters must share materials or atlases to stay cheap.
5. **Export a cluster.** Put the cluster in its own collection with its own Collection Exporter and a pivot at its base, named like `SM_MillAlley_Cluster_WallBase_01`.
6. **Make two or three versions.** The same pieces rearranged give variety without new assets.

### Common mistakes

- Debris spread evenly like confetti. It gathers where wind, water and people push it.
- Joined clusters with many materials. Each material is a draw call; use atlased props.
- Floating pieces. Check from a low angle with a light from the side.

### Drill: one wall-base debris cluster (40 minutes)

Build a 2 m long debris cluster for the base of a wall from your planks, bricks, rubble, litter and leaves, and export it as one mesh.

**Pass when:** it reads as a natural pile from Camera 1's height, nothing floats, it uses no more than three materials, and it imports into UE5 with its pivot at ground level.

### Use it on Mill Alley

Today is ground dressing: rubble edges along wall bases and debris following drainage and foot traffic. “Done when” is matching the photo’s density. Place your clusters along the walls first, then fill in by hand.

## Tue Dec 29 · Composition: thirds, leading lines and framing

**Goal:** use composition rules and Blender’s camera guides to make shots that lead the eye to the focal point.

**Time:** about 40 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Understanding Composition — Blender Guru (Andrew Price) · About 23 min of composition theory: thirds, leading lines, framing. The theory doesn’t depend on the Blender version (2014)
- **Video:** Frame Your Renders Better with Camera Composition Guides - Blender Quick Tips — 247 Creative Secrets · Short 2025 how-to for turning the guides on in current Blender
- **Manual:** Cameras — Camera Properties > Viewport Display > Composition Guides (Thirds, Center/Diagonal, Golden ratio/triangles, Harmony triangles). In camera view, turn them on or off with Overlays > Camera Guides

### Steps

1. **Composition guides.** Select a camera > Object Data properties > Viewport Display > *Composition Guides*. Turn on *Thirds*, and try *Golden* and *Diagonal*. They draw over the camera view. In camera view you can also toggle them from Overlays > *Camera Guides*.
2. **Rule of thirds.** Put the focal point (the bright far opening) on or near a thirds intersection, not dead centre, unless you want a strong symmetrical corridor shot (the reference photo is close to central, and that’s fine for Camera 1).
3. **Leading lines.** The walls’ edges, ground joints, the handrail, planks and cables should point toward the focal point. Rotate props so their long edges join that flow.
4. **Framing.** Dark foreground shapes at the edges (a wall, a hanging branch, the skybridge) frame the bright centre and add depth.
5. **Values guide the eye.** The eye goes to the brightest, highest-contrast area first. Keep the focal point the lightest area, and keep busy detail away from frame edges.
6. **Foreground, midground, background.** Each shot should have something in all three layers.
7. **Try many quickly.** Duplicate the camera (Shift+D), try a new angle, and render a quick EEVEE or viewport image. Compare them side by side.

### Common mistakes

- Everything the same size and brightness. Then nothing leads.
- Props pointing out of frame. Lines that run off the edge pull the eye out of the image.
- A camera too high. Eye level (1.5–1.7 m) or lower usually feels more immersive in an alley.

### Drill: three alternate compositions of your greybox (40 minutes)

Using the Mill Alley greybox or current scene in Blender, set up three new cameras (low angle up at the skybridge, light well close-up, reverse shot from the far opening), each using thirds and leading lines.

**Pass when:** each render has a clear focal point on or near a thirds point, at least two leading lines aimed at it, and something in the foreground, midground and background.

### Use it on Mill Alley

Today you place P1 props in Camera 1 for composition (plywood, planks, branches, cables). “Done when” is leading lines pointing to the far opening; check Camera 1 against the same thirds lines in UE5 (a screenshot with a thirds grid drawn over it works). The drill’s three angles become the alternate cameras.

## Fri Jan 1 · Buffer (New Year’s Day)

**Goal:** no lesson today.

### Steps

1. Rest. If you want to do something, save a dated progress screenshot of Camera 1 for the process sheet.

### Use it on Mill Alley

The Mill Alley plan is also on buffer today.

## Sat Jan 2 · Geometry Nodes scattering on chosen areas: weeds in cracks

**Goal:** scatter weeds only where you paint, such as in ground cracks and along wall bases.

**Time:** about 50 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Video:** Scattering Objects by WEIGHT PAINT with Geometry Nodes in Blender! — The CG Essentials · Restrict a scatter to painted or selected areas: exactly the crack-weed setup (2023, Blender 3.4+ nodes; same in 5.x)
- **Video:** Blender 4.2 - An intro to Geometry Nodes and Scattering — Nino · Optional refresher on the basic scatter graph in 4.2+ (Oct 2024)
- **Manual:** Distribute Points on Faces Node — Use the Selection input with a named attribute or vertex group so points land only on the crack/wall-base faces
- **Manual:** Instance on Points Node — Instance the weed meshes; add random rotation and scale

### Steps

1. **Paint where weeds grow.** On a ground tile, open Object Data > Vertex Groups, click + and name it `Weeds`. In *Weight Paint* mode, paint red where weeds should grow (cracks, wall edges, around drains) and leave the rest blue.
2. **Scatter by weight.** Add a Geometry Nodes modifier. *Distribute Points on Faces* (Random) has a *Density Factor* input. Add a *Named Attribute* node (type *Float*, name `Weeds`) and plug it into Density Factor. Points now appear only where you painted, thicker where the paint is stronger.
3. **Instance weeds.** *Instance on Points* with your F03 weed clumps from a *Collection Info* node (*Pick Instance* on), random rotation around Z, random scale.
4. **Keep the ground.** Add *Join Geometry* so the output has both the ground and the weeds.
5. **Scale by weight too.** Feed the same Named Attribute into a *Map Range* and then into Scale, so weeds are bigger in the middle of a painted patch and smaller at its edges.
6. **Games vs Blender.** In UE5 you’ll place most weeds with the Foliage tool. Use this scatter for baked clusters (a weedy crack strip you export as one mesh) and for planning density in Blender renders.

### Common mistakes

- Weeds everywhere. The Named Attribute name must match the vertex group exactly.
- Weeds sunk or floating. The instance’s origin must be at its base.
- Uniform density. Real weeds cluster where water and soil collect.

### Drill: a crack-weed scatter on one ground tile (30 minutes)

Paint a `Weeds` group along two cracks and a wall edge on a D01 tile and scatter weed clumps there with Geometry Nodes.

**Pass when:** weeds appear only along the painted areas, bigger in the middle of patches and smaller at edges, and changing the paint updates the scatter.

### Use it on Mill Alley

Today is foliage placement: weeds in cracks, moss on ledges and steps, buddleia in mortar, and Review 11. “Done when” is foreground to background reading clearly; use your drill as a guide for where UE5’s foliage tool should paint densest.

## Sun Jan 3 · Buffer: catch up on Weeks 8 to 11

**Goal:** close the gaps from Weeks 8 to 11.

**Time:** as long as you need; no new lesson.

### Steps

1. **List your gaps.** Any drill from Weeks 8 to 11 not ticked, or ticked without meeting its *Pass when*.
2. **Priority order.** Redo these first if they’re on your list, since the final renders depend on them:
   - The prop sheet render (Week 8, Sun).
   - The foliage sheet (Week 9, Sun).
   - Three barrel variants from one mask (Week 10, Sun).
   - A facade with enough vertices for smooth paint (Week 10, Sat).
3. **Write one line per drill** in the tracker’s notes: what failed and what fixed it.

### Use it on Mill Alley

The Mill Alley plan is on buffer today too. Use it for the plan first if you’re behind.

# Week 12 · Jan 4 to 10 · Presentation scenes and cleanup

This week's goal is to light small showcase scenes well, model wet ground details, and clean every mesh before the final lighting and renders.

## Mon Jan 4 · Presentation lighting: HDRI, three-point, EEVEE vs Cycles

**Goal:** light a small vignette so it looks like a finished portfolio piece.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** The ULTIMATE GUIDE to HDRI Lighting in Blender! — The CG Essentials · Feb 2024. Watch 4:29-13:21 (add, rotate, light with an HDRI); 13:21+ dome setup is optional
- **Video:** Lighting for Beginners! (Blender Tutorial) — Ryan King Art · Watch 35:44-43:57, the three-point lighting setup. Skip 21:59 'Bake Eevee Lighting' (old EEVEE; not needed in 4.2+)
- **Manual:** World Environment — HDRI via Environment Texture in the World
- **Manual:** Light Objects — Area/spot/point lights for the key, fill and rim

### Steps

1. **HDRI for ambient light.** World Properties > Surface > Color > *Environment Texture* and open an HDRI from Poly Haven. Add a *Mapping* node (Ctrl+T with Node Wrangler) to rotate it with Z rotation so the brightest part of the sky lights your subject from the side.
2. **Hide the HDRI if needed.** To light with the HDRI but show a plain background, go to Render > Film and tick *Transparent*, then put a colour behind it in compositing, or use a backdrop sweep.
3. **Add three-point lights on top.** Key, fill and rim area lights (Week 6) give control the HDRI can't. The HDRI gives soft fill and reflections; the lights shape the subject.
4. **EEVEE or Cycles.** EEVEE renders in seconds: great for iterating and most prop shots. Cycles traces light properly: better for soft bounce light, glass and fine sculpt detail. Set up in EEVEE, check final frames in Cycles.
5. **Cycles settings.** Render Properties: Samples around 256 with *Denoise* on. Light Paths: defaults are fine.
6. **Depth of field.** Camera properties > *Depth of Field*: pick a *Focus Object* (the hero prop) and an F-Stop around 2.8–5.6 for gentle background blur. Don't overdo it; environment art reviewers want to see the work.
7. **Compare.** Render the same frame in EEVEE and Cycles and note the differences.

### Common mistakes

- A strong HDRI and strong lights fighting. Turn the HDRI's Strength down when the lights take over.
- Over-blurred backgrounds. Keep the subject and its surroundings readable.
- Judging lighting in Material Preview. Use Rendered view.

### Drill: a small prop showcase vignette (40 minutes)

On a 2 × 2 m patch of your ground tile, arrange a few props (barrel, planks, rubble, weeds) and light it with an HDRI plus a key and rim light. Render it in EEVEE and Cycles.

**Pass when:** the props separate from the background, the rim light outlines the silhouettes, nothing is pure black or pure white, and you can say which renderer you'd use for the final shot and why.

### Use it on Mill Alley

Today you dress the alternate cameras with P2 props, glass, litter and paper ("no empty corners"). This vignette becomes the separate prop showcase scene.

## Tue Jan 5 · Puddle inserts and wet surfaces

**Goal:** model a shallow puddle insert that sits flush with the ground tile and reads as water.

**Time:** about 40 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Read:** Blender Tutorial: Photorealistic Puddles — Blender Guru (Andrew Price) · Mar 2023. The wet-surface look (roughness and darkening masks, a flat water layer); use it for the shading half of the lesson
- **Manual:** Inset Faces — Inset the puddle outline in D01, then lower the inner faces slightly. Keep the border verts shared so the insert sits flush
- **Manual:** Shrinkwrap Modifier — Snap a separate puddle insert to the D01 surface so its edges match exactly

### Steps

1. **Where puddles form.** In low points: dips in the ground tile (the dip you made in Week 5), along wall bases and in wheel ruts.
2. **The outline.** D06 is about 150 × 100 cm. Add a Circle with 24–32 vertices, fill it (F), and move edge vertices with proportional editing into an irregular, natural shape.
3. **Flush edges.** The outline must sit exactly on the ground surface. Use snapping to *Face* (*Project Individual Elements* on) and move the edge vertices onto D01 so there's no gap or step.
4. **A shallow dip.** Inset the inner faces a few times (I) and lower them 2–3 cm with proportional editing. Puddles are shallow.
5. **The water surface.** A separate flat plane cut to the same outline, set just below the rim (about 1 cm). Give it a dark base colour, Roughness about 0.02 and a subtle ripple normal. In UE5 it reflects the sky and walls via Lumen.
6. **Wet edges.** The ground around a puddle is wet too: a ring of darker, glossier surface. Use vertex paint blue (the wet channel) or an X07 decal around the insert.
7. **Pivot and export.** Pivot at the centre of the base, name `SM_MillAlley_D_PuddleInsert_150x100`.

### Common mistakes

- Puddles floating or sunk too deep. Snap the rim to the ground surface.
- Perfect ovals. Real puddle edges are irregular.
- No wet halo. Without it, the water looks pasted on.

### Drill: one D06 puddle insert (30 minutes)

Model a puddle insert into the dip on your D01 tile, with a water plane and a wet halo.

**Pass when:** the rim meets the ground with no visible step from Camera 1's height, the water plane reflects the sky in Rendered view, and the shape is irregular.

### Use it on Mill Alley

Today is puddles and wetness: X07 decals, D06 inserts and wet roughness on the ground. "Done when" is puddles reflecting the sky in UE5.

## Fri Jan 8 · Buffer

**Goal:** no lesson today.

### Steps

1. Use the time for the Mill Alley plan, or redo any unfinished drill from Weeks 8 to 12.

### Use it on Mill Alley

The Mill Alley plan is also on buffer today.

## Sat Jan 9 · Mesh cleanup

**Goal:** find and fix the hidden mesh problems that cause shading errors, bad bakes and lighting bugs in UE5.

**Time:** about 40 minutes for the lesson; the drill runs as long as your kit needs.

### Watch and read first

- **Video:** Blender Tutorial - How to Merge By Distance, Clean Up Your Mesh — GRAPHICUE · Short Oct 2024 demo of Merge by Distance
- **Video:** Healing Non-Manifold Objects with Blender — GridSpace · Finding and fixing non-manifold edges (made for 3D printing, but the method is the same; 2021)
- **Manual:** Clean Up — Delete Loose, Degenerate Dissolve, Fill Holes. Merge by Distance is in the Merge page
- **Manual:** Merge — Merge by Distance (M > By Distance)
- **Manual:** Editing Normals — Recalculate Outside (Shift+N)
- **Manual:** Select All by Trait — Select > Select All by Trait > Non Manifold / Loose Geometry / Interior Faces

### Steps

1. **Merge by Distance.** In Edit Mode, select all and use Mesh > Clean Up > *Merge by Distance* (or M > *By Distance*). It welds duplicate vertices, a common result of extrudes you cancelled. The report at the bottom tells you how many were removed.
2. **Recalculate normals.** Select all and press Shift+N (*Recalculate Outside*). Then check with the Face Orientation overlay: everything blue.
3. **Non-manifold geometry.** Select > Select All by Trait > *Non Manifold*. It finds holes, edges shared by more than two faces, and loose edges. Holes on visible surfaces must be filled; edges at the back of a module that touches a wall can be fine.
4. **Loose geometry.** Mesh > Clean Up > *Delete Loose* removes stray vertices and edges that aren't part of any face.
5. **Degenerate faces.** Mesh > Clean Up > *Degenerate Dissolve* removes zero-area faces that break normals and bakes.
6. **Re-check after.** Re-run the snap test, re-check texel density and triangle counts.
7. **Make a checklist file.** List every kit file and tick each step for each one. Then reimport the fixed meshes into UE5 (same names, same pivots).

### Common mistakes

- Merge by Distance with too high a distance. It can collapse small bevels; keep it at the default (0.0001 m).
- Recalculating normals on intentionally open meshes. Check visually after.
- Fixing in UE5. Fix the source in Blender and reimport, or the next export brings the bug back.

### Drill: clean every kit file (open-ended)

Run steps 1 to 5 on every module, hero and prop file and reimport them.

**Pass when:** every file's Merge by Distance reports 0 removed on a second run, Face Orientation is all blue, there's no non-manifold geometry on visible surfaces, and every reimported asset still sits in the same place in UE5.

### Use it on Mill Alley

Today is the full walkthrough: fix floating objects, seams and repeated props, and Review 12. Your drill is half of that fix list. For the standard check, have the separate prop showcase scene ready, as Courtyard does.

## Sun Jan 10 · Buffer

**Goal:** no lesson today.

### Steps

1. Use the time for the Mill Alley plan, or keep going with the cleanup drill if your kit is large.

### Use it on Mill Alley

The Mill Alley plan is also on buffer today. Lighting starts tomorrow, so the scene should be clean.

# Week 13 · Jan 11 to 17 · Light and value

This week's goal is to understand light well enough to light the alley in UE5: sun and sky, exposure and colour temperature, fog and light shafts, values in greyscale, and mood changes.

## Mon Jan 11 · Light basics that carry into UE5

**Goal:** learn the few lighting ideas every engine shares, and test them on Camera 1 in Blender.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Lighting for Beginners! (Blender Tutorial) — Ryan King Art · Watch 6:32-8:21 (sun light) and 24:54-27:57 (world, Sky Texture)
- **Manual:** Light Objects — Sun: Strength in W/m² and Angle (shadow softness); Temperature (blackbody Kelvin) for warm/cool light
- **Manual:** Sky Texture Node — Physical sky to pair with the sun, the closest match to UE5 Sky Atmosphere
- **Manual:** Displays and Views — Exposure (stops) and the AgX view
- **Unreal docs:** Sky Atmosphere Component in Unreal Engine — The UE5 side of sun and sky
- **Unreal docs:** Auto Exposure in Unreal Engine — Lock exposure (Manual) so the UE5 values match your Blender study

### Steps

1. **Two light sources outdoors.** The *sun* is one strong directional light with sharp shadows. The *sky* is a big, soft, blue-ish light from everywhere above. Overcast weather moves almost all the light from the sun to the sky.
2. **In Blender.** Sun lamp = UE5's *Directional Light*. World colour or HDRI = UE5's *Sky Light* (plus *Sky Atmosphere* for a physical sky). The same thinking works in both.
3. **Colour temperature.** Light colour is measured in Kelvin: candle about 1,900 K, tungsten bulb about 3,000 K, midday sun about 5,500 K, overcast sky about 6,500–7,500 K (cooler, bluer). Blender 5.2's lights can take their colour from a *Temperature* value in Kelvin, just like UE5's lights.
4. **Warm vs cool.** Where the sun hits is warm; shadows lit by the sky are cool. That contrast is most of what makes lighting feel real. In the alley, the far opening is the bright, slightly warm area and the walls sit in cool shade.
5. **Exposure.** Like a real camera, exposure sets how bright the whole image is. Keep lights at realistic ratios (sun much brighter than sky) and adjust *Exposure* in Color Management. In UE5, lock exposure (Post Process Volume > Exposure > Metering Mode *Manual*) so it doesn't drift while you work.
6. **Bounce light.** Light bouncing off walls and the ground fills the shadows. Cycles does this naturally; UE5's Lumen does it in real time. In EEVEE, enable ray tracing in Render Properties for better bounce.
7. **Use Camera 1.** Every study this week is through your camera-match camera, so you can compare directly with the photo.

### Common mistakes

- Fixing a dark scene by brightening every light. Change exposure instead, and keep the ratios.
- Everything the same colour temperature. Without warm/cool contrast the image looks flat.
- Exposure that keeps changing in UE5. That's auto-exposure; set it to Manual while lighting.

### Drill: an EEVEE overcast study of Camera 1 (40 minutes)

In your camera-match file (or a Blender export of the current scene), light Camera 1 with a cool sky (HDRI or world colour), a weak warm sun from the far opening and exposure set by eye against the photo.

**Pass when:** placed beside the photo, the overall brightness, the bright far end, and the cool shaded walls match closely, and the shadow areas aren't black.

### Use it on Mill Alley

Today in UE5 you set up the overcast sky, a soft directional light, the skylight and Lumen, and lock exposure. "Done when" is a flat, cool base matching the photo; your Blender study shows the target values.

## Tue Jan 12 · Volumetrics: fog and light shafts

**Goal:** add atmospheric depth and a soft light shaft, and learn what each setting does.

**Time:** about 50 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** How to Create Volumetric Lighting in Blender (Eevee & Cycles) — Shift4cube · Dec 2024, so it covers EEVEE Next (4.2+) volumetrics
- **Manual:** Volumes (EEVEE render settings) — Render > Volumes: Resolution, Steps, Distribution, Max Depth, Custom Range (use with a World volume to cut noise)
- **Manual:** Light Settings (EEVEE) — Light > Influence > Volume Scatter multiplier; Jitter is for soft surface shadows and is not needed for shafts
- **Manual:** Sampling (EEVEE) — Render > Sampling > Shadows > Volumetric Shadows (+ Steps): needed for the shaft occlusion
- **Manual:** Principled Volume — Plug into World > Volume (low density) or into a box volume at the far opening
- **Unreal docs:** Volumetric Fog in Unreal Engine — Exponential Height Fog > Volumetric Fog; light shafts from a light's Volumetric Scattering Intensity

### Steps

1. **Why fog helps.** Air between camera and distant objects lightens and softens them, which separates foreground, midground and background. Even a little fog adds a lot of depth.
2. **World fog.** World Properties > *Volume*: add a *Principled Volume* with a very low Density (0.005–0.02). Too much and everything turns milky.
3. **Local fog.** For more control, add a large cube around the far end of the alley, give it a material with only a *Principled Volume* plugged into the *Volume* output (nothing in Surface), and set its density. Fog sits where you want it.
4. **Light shafts.** Shafts appear where a light passes through fog and is partly blocked: here, sunlight through the far opening and past the skybridge. The light needs shadows on, and the fog must be where the light passes.
5. **EEVEE settings.** Shafts come from *Volumetric Shadows*, under Render Properties > Sampling > Shadows; turn it on. Render Properties > *Volumes* sets fog quality (Resolution, Steps); with a World volume, tick *Custom Range* and set an end distance to cut noise. On the sun, Influence > *Volume Scatter* sets how strongly it shows in the fog. If shafts still look blocky, render the study in Cycles.
6. **The UE5 version.** *Exponential Height Fog* with *Volumetric Fog* on, and the Directional Light's *Volumetric Scattering Intensity* controls how much each light shows in the fog.

### Common mistakes

- Fog so dense the scene looks like smoke. Start very low and raise slowly.
- No shafts at all. Volumetric Shadows is off, or nothing blocks the light.
- Fog on the near walls. Keep the foreground clear and let fog build toward the far end.

### Drill: a fog study with a light shaft at the far opening (40 minutes)

In Camera 1, add low world fog plus a denser fog volume at the far end, and angle the sun so a soft shaft falls through the far opening.

**Pass when:** the far end is lighter and softer than the near walls, a shaft is visible but subtle, and the foreground stays crisp.

### Use it on Mill Alley

Today in UE5 you add height fog and volumetrics, a soft shaft at the far opening, and bounce and fill lights. “Done when” is depth reading in three layers, exactly what your study shows.

## Fri Jan 15 · Colour management and reading values

**Goal:** understand view transforms and exposure, and check your image in greyscale to see if its values work.

**Time:** about 40 minutes for the lesson and 30 for the drill.

### Watch and read first

- **Read:** Color Management (Blender Fundamentals 4.5 LTS) — Blender Studio (Beau Gerbrands) · Free official lesson; watch it all
- **Read:** The Secret to Rendering Vibrant Colors with AgX in Blender is the Raw Workflow — CG Cookie (Jonathan Lampel) · Optional deeper read on AgX and grading (Apr 2024)
- **Manual:** Color Management — Overview and color spaces
- **Manual:** Displays and Views — View transforms: AgX (default), Khronos PBR Neutral, False Color (exposure heat map); Look, Exposure, Gamma
- **Manual:** RGB to BW Node — For a value-only render, put this compositor node (now under Color nodes) before the Group Output. For a quick exposure check, use the False Color view transform

### Steps

1. **View transform.** Render Properties > Color Management > *View Transform*. *AgX* (the default) handles bright light gracefully, rolling highlights off instead of clipping them. *Standard* clips and looks harsher. Stay on AgX.
2. **Looks.** The *Look* menu adds contrast presets (AgX - Punchy, Base Contrast, and others). Try them, but make big changes with lighting first.
3. **Exposure and Gamma.** Exposure brightens or darkens everything evenly. Gamma changes the midtones; leave it at 1 unless you have a reason.
4. **What “values” are.** A value is how light or dark something is, ignoring colour. A good environment image has a clear value structure: the focal point lightest, the frame darker, and maybe three or four main value groups.
5. **Greyscale check in Blender.** In the *Compositing* workspace, add a node tree if there isn’t one, and put an *RGB to BW* node (Add > Color) between the *Render Layers* node and the *Group Output* node (Blender 5 replaced the old Composite node with Group Output). Render (F12): the result is greyscale. Remove it afterwards.
6. **Or outside Blender.** Desaturate a saved render in any image editor. Same result.

7. **Squint.** Squint at the greyscale image: if the far opening doesn't pop as the brightest area, change lighting or materials until it does.

### Common mistakes

- Saturated colour hiding weak values. In greyscale, colourful-looking images can turn to mush.
- Using a Look to fix lighting. Fix it in the lights; Looks are a final touch.
- Comparing renders with different view transforms. Keep AgX for all.

### Drill: a value-only render of Camera 1 (30 minutes)

Render Camera 1 in greyscale and mark three value groups (light, mid, dark) on a printout or a copy of the image.

**Pass when:** the far opening is clearly the lightest group, the two walls separate from each other and from the ground, and no large area is a muddy mid-grey with no structure.

### Use it on Mill Alley

Today in UE5 you do post process (grade toward the photo, light vignette, grain) and tune reflections on glass and puddles. Make the same greyscale check on UE5's Camera 1 screenshot. For the render list, keep this as the value (greyscale) shot.

## Sat Jan 16 · Lighting moods

**Goal:** light the same camera three ways to learn how mood comes from light.

**Time:** about 30 minutes for the lesson and 60 for the drill.

### Watch and read first

- **Video:** Lighting The Same Scene For Day AND Night — Brady Bessette · Exactly the drill: one scene, several moods (2022)
- **Video:** 3 Easy Lighting Setups | Blender Tutorial — Max Hay · Optional; Nov 2025
- **Manual:** World Environment — Swap the world per mood
- **Manual:** Sky Texture Node — Sun elevation/rotation for morning vs overcast; night = low sky strength plus practical lights

### Steps

1. **One file, several setups.** Scene menu (top right) > *New Scene* > *Linked Copy*. The new scene shares all the objects but can have its own world, lights and render settings. Make one scene per mood.
2. **Morning.** A low, warm sun (3,500–4,500 K) raking along the alley, long shadows, cool blue sky fill, light fog catching the sunlight.
3. **Overcast.** Weak or no sun, a strong neutral-cool sky, soft shadows, low contrast. This is the reference photo's mood.
4. **Night.** No sun, a dark blue-violet sky at low strength, and practical lights: a warm point light in the P18 wall lamp, maybe a cool light from a window. Exposure goes up a little, but keep the night dark.
5. **Change only light.** Don't move the camera or props between moods; the comparison teaches you what light alone does.
6. **Compare in greyscale.** Run each mood through the value check: the focal point should still lead in all three.

### Common mistakes

- Night that's just "darker day". Night has different colours and light sources.
- Morning that's too orange. Warm sun, but sky light stays blue.
- Lights duplicated rather than linked. Use Linked Copy scenes so models stay in sync.

### Drill: three mood studies (60 minutes)

Light Camera 1 as morning, overcast and night in three linked-copy scenes and render each.

**Pass when:** the three read as clearly different times or weather, the camera and props are identical in all three, and the focal point still reads first in each.

### Use it on Mill Alley

Today in UE5 you light the alternate cameras so each has a clear focal point. Pick one mood (for example dusk or night) as the alternative lighting mood for the render list, and rebuild it in UE5 later this week.

## Sun Jan 17 · Review: studies vs UE5

**Goal:** compare your Blender studies with UE5's renders, and get outside critique.

**Time:** about 60 minutes.

### Steps

1. **Side by side.** Put the photo, your Blender overcast study and the UE5 Camera 1 render in one image.
2. **Compare in greyscale.** Values first, then colour temperature, then fog depth.
3. **List differences.** For each, write whether UE5 or the study is closer to the photo and what you'll change.
4. **Share for critique.** Send the lit renders to peers or a mentor (Discord art communities, Polycount, a teacher). Ask specific questions: "Does the focal point read?", "Is the fog too heavy?"

### Use it on Mill Alley

Today is Review 13: share lit renders for critique and collect notes. For the standard check, have a value shot and one alternative lighting mood ready, as Greenhouse, Indian street and Windmill show.

# Week 14 · Jan 18 to 24 · Game-ready polish

This week's goal is the professional finish: fix critique notes without breaking the scene, add LODs and collision where they matter, and audit the whole kit for density, triangles and naming.

## Mon Jan 18 · Fixing assets after critique without breaking UE5

**Goal:** apply critique notes to meshes and update them in UE5 with nothing moving or duplicating.

**Time:** about 30 minutes for the lesson; the drill depends on your notes.

### Watch and read first

- **Video:** Blender to Unreal Engine 5 Via FBX - Keep Your Pivot Points Intact — Miguel De Andrade · Jul 2024; pivot handling so reimports don't shift placed actors
- **Manual:** Object Origin — Keep pivots identical (bottom-left-back) when fixing meshes
- **Manual:** FBX (Legacy) — Re-export with the same object name and settings so UE5 reimports onto the same asset
- **Unreal docs:** FBX Static Mesh Pipeline in Unreal Engine — Static mesh import/reimport rules
- **Unreal docs:** Reimporting Assets Automatically in Unreal Engine — Editor Preferences > Loading & Saving > Auto Reimport > Monitor Content Directories. Manual route: right-click asset > Reimport

### Steps

1. **Sort the notes.** For each critique note, write which asset it touches, whether it's a mesh, material or placement fix, and whether it's worth doing in the time left. Skipping a note on purpose is fine; ignoring it isn't.
2. **The three rules.** When you change a mesh: keep the same name, keep the same pivot, export with the same preset to the same file (Week 2).
3. **Silhouette fixes.** Changes to the outline (a straighter roofline, a deeper sill) are worth geometry. Surface fixes are often better done with decals or vertex paint in UE5.
4. **Variants instead of edits.** If a module is used in many places but one placement needs a change, export a new variant (_v2) rather than changing the original everywhere.
5. **Reimport and check.** Reimport in UE5 (right-click > Reimport). Look at every place the asset is used, including alternate cameras.
6. **Log it.** In the tracker's notes, list each note with "fixed" or "skipped because…".

### Common mistakes

- Renaming during a fix. UE5 imports a new asset and the old one stays in the level.
- Moving the pivot to "fix" a placement. Every copy shifts. Move the placed actor in UE5 instead.
- Fixing everything. Prioritize notes that affect Camera 1 and the main renders.

### Drill: fix two critiqued meshes and reimport them (time varies)

Pick two notes that need mesh changes, fix them in Blender and reimport.

**Pass when:** both fixes show in every placement, nothing moved, and no duplicate assets appear in the Content Browser.

### Use it on Mill Alley

Today you apply critique notes, each fixed or consciously skipped. This drill is the mesh half of it.

# Tue Jan 19 · LODs: levels of detail

**Goal:** make lower-detail versions of a prop that UE5 swaps in at a distance, and know when Nanite makes them unnecessary.

**Time:** about 40 minutes for the lesson and 30 for the drill.

## Watch and read first

- **Video:** Export LODs from Blender to UE5 — CG Krab · Jul 2023; the Blender-side LOD setup for UE5
- **Manual:** Decimate Modifier — Collapse ratio about 0.5 for LOD1 and 0.25 for LOD2; apply on copies
- **Unreal docs:** Importing Static Mesh LODs Using FBX — LOD Group in the FBX, then Import Options > Mesh > Advanced > Import LODs
- **Unreal docs:** Creating and Using LODs in Unreal Engine — Static Mesh Editor > Details > LOD Settings > LOD Import > 'Import LOD Level 1' (Reimport LOD Level N to replace)

## Steps

1. **What LODs are.** LOD0 is the full mesh. LOD1 and LOD2 have fewer triangles and swap in as the object gets smaller on screen. Viewers never notice; the frame rate does.
2. **Nanite first.** UE5's Nanite handles detail automatically for opaque static meshes, so Nanite-enabled kit pieces and heroes don't need hand-made LODs. (Walls you vertex paint per copy in UE5 stay non-Nanite, since per-copy paint doesn't show on Nanite meshes; give those LODs if they're heavy.) LODs still matter for meshes that don't use Nanite in your project (often foliage cards, translucent glass, and some masked materials, depending on your UE5 version and settings).
3. **Make LOD1.** Duplicate the mesh, add a *Decimate* modifier (Collapse), Ratio about 0.5. Watch the silhouette and UV seams; raise the ratio if the outline breaks. Apply it.
4. **Make LOD2.** Duplicate again, Ratio about 0.25. Small props can drop details completely at LOD2 (bolts, thin slats).
5. **Keep UVs and materials.** Decimate keeps UVs on most meshes; check with the UV Grid. All LODs use the same material and textures.
6. **Name and export.** `SM_MillAlley_P13_Barrel_LOD1`, `_LOD2`, each exported as its own FBX with the same pivot as LOD0.
7. **Import in UE5.** Open the LOD0 mesh in the Static Mesh Editor. In Details > *LOD Settings*, set the number of LODs, then use *LOD Import* to import each LOD file. Use the viewport's LOD dropdown to check each.
8. **Or let UE5 do it.** The Static Mesh Editor can also generate LODs automatically (*LOD Settings* > reduction). Hand-made LODs give you control; auto LODs are fine for simple props.

## Common mistakes

- LODs that pop visibly. The silhouette changed too much; raise the ratio.
- LODs with a different pivot. They jump when they swap.
- Making LODs for Nanite meshes. Check *Enable Nanite* first; if it's on, you don't need them.

## Drill: LODs for one prop (30 minutes)

Make LOD1 and LOD2 for your barrel or a plank pile and import them into UE5.

**Pass when:** LOD1 is about half and LOD2 about a quarter of LOD0's triangles, the silhouette holds at the distance each switches in, and UE5 shows all three in the LOD dropdown.

## Use it on Mill Alley

Today is material and asset polish: tiling break-up, roughness ranges and the greyscale value check. Add LODs only to the non-Nanite meshes you find while polishing.

# Fri Jan 22 · Collision

**Goal:** give meshes simple collision so the alley is walkable, and know when to use simple vs complex collision.

**Time:** about 40 minutes for the lesson and 30 for the drill.

## Watch and read first

- **Video:** Blender to Unreal: Custom Collision for UE5 (E02) — CG Krab · About 15 min. 9:10 proxy mesh, 10:42 naming, 12:15 export/import, 14:10 UE settings
- **Manual:** Convex Hull — Build UCX_ pieces from a duplicate of the mesh; split concave shapes into several hulls
- **Unreal docs:** FBX Static Mesh Pipeline in Unreal Engine — Collision section: UCX_/UBX_/UCP_/USP_ + [RenderMeshName]_##, same FBX, convex only, no overlaps
- **Unreal docs:** Simple versus Complex Collision in Unreal Engine — Collision Complexity options: when to use simple vs complex-as-simple

## Steps

1. **What collision is.** An invisible, simple shape the physics and player use instead of the real mesh. Simple shapes are fast; the full mesh as collision is slow.
2. **Simple collision from Blender.** Model low-poly convex shapes around the mesh (boxes mostly). Name each `UCX_<MeshName>_00`, `_01`… exactly matching the mesh name, and export them in the same FBX as the mesh. UE5 recognises them as collision.
3. **Convex only.** Each UCX shape must be convex (no dents or holes), and several shapes on one mesh shouldn't overlap. An L-shaped wall needs two boxes side by side, not one L.
4. **Other prefixes.** `UBX_` for boxes, `USP_` for spheres, `UCP_` for capsules, with the same naming pattern.
5. **Or make it in UE5.** The Static Mesh Editor's *Collision* menu adds boxes, spheres or auto convex hulls. Quick for simple props.
6. **Simple vs complex.** *Simple* = your UCX shapes (used for walking and physics). *Complex* = the full mesh (used for precise traces). For walls and ground, simple boxes are plenty. Use *Use Complex Collision as Simple* only for rare cases like an uneven ground you walk on.
7. **What needs it.** Walls, ground, stairs and big props the player can bump. The skybridge, out of reach, needs only a couple of boxes. Tiny debris and foliage need none (turn collision off for them).
8. **Check it.** In the Static Mesh Editor, Show > *Simple Collision*. In the level, walk the alley in Play mode.

## Common mistakes

- UCX names that don't match the mesh name. UE5 ignores them.
- Concave collision shapes. They're turned into wrong shapes or ignored. Split into convex parts.
- Collision on every pebble. It costs performance and snags the player.

## Drill: collision for the skybridge and two props (30 minutes)

Make UCX collision for the skybridge (two or three boxes), a plank pile and the barrel, and check them in UE5.

**Pass when:** Show > Simple Collision outlines each mesh closely with convex shapes, the names match, and in Play mode you can't walk through the props.

## Use it on Mill Alley

Today is optimization: Nanite and LODs, texture streaming, collision, and profiling with `stat unit` and the GPU Visualizer. "Done when" is a stable frame rate in a walkthrough.

## Sat Jan 23 · Kit audit and the kit layout

**Goal:** check every module against the project's rules, then lay the kit out with labels and dimensions for the kit sheet.

**Time:** about 30 minutes for the lesson; the drill runs as long as the kit needs.

### Watch and read first

- **Video:** Understanding Texel Density for Environments & Props — Hugh Chew · Jan 2026. Watch 0:39-19:28 (theory, real-world reference, checking it in UE5); the 19:29+ demo is in Maya
- **Read:** Texel Density Checker (extension) — IvanVostrikov / extensions.blender.org · Supports 4.2+, v2026.1.1 (Jul 2026). Install from Get Extensions
- **Manual:** Viewport Overlays — Statistics overlay for triangle counts
- **Manual:** Text Objects: Introduction — Text labels for the kit grid
- **Manual:** Measure — Dimension lines on the kit sheet

### Steps

1. **Texel density.** Open each module file and run Texel Density Checker's *Calculate TD* on every visible face. Near surfaces: 10.24 px/cm; upper floors and background: 5.12.
2. **Triangle counts.** Read each module's count from Statistics and compare it with the plan's budget table.
3. **Naming.** Every asset follows `SM_MillAlley_<Family>_<Piece>_<W>x<H>`; textures `T_…`; materials `M_…` or `MI_…`.
4. **A spreadsheet.** Record density, triangles and naming in the tracker spreadsheet, one row per module. Fix anything out of range.
5. **Lay out the kit.** In a new Blender scene (or UE5 level), place every module on a grid in family rows (A, B, C, D), with even spacing.
6. **Labels.** For each piece, a Text object (Shift+A > Text) with its ID and size, e.g. "A01 · 400 × 400", rotated to face the camera. Add a 180 cm figure at the end of a row for scale.
7. **Assembled examples.** Next to the rows, build two walls from the same pieces arranged differently to show the kit's flexibility.

### Common mistakes

- Auditing from memory. Check every file; small mistakes hide in modules you made early.
- Labels too small. They must read in a downsized ArtStation image.
- Missing the scale figure. Viewers judge size from people.

### Drill: the full kit layout (open-ended)

Audit every module and build the labelled kit layout with a scale figure and two assembled walls.

**Pass when:** every module meets density, budget and naming (or has a written reason), every piece has a readable ID and size label, and the layout includes a 180 cm figure.

### Use it on Mill Alley

Today you audit texel density, wireframes and naming, and build a kit breakdown level with every module laid out. The drill and the plan's task are the same job; do the layout wherever you'll shoot it (UE5 matches the final renders best).

## Sun Jan 24 · Review: final kit audit

**Goal:** sign off the kit before the final renders.

**Time:** about 45 minutes.

## Steps

1. **Run the audit sheet.** Every row green, or with a written reason.
2. **Shoot a test.** Render the kit layout once, at final resolution, and check that labels and wireframes read at ArtStation's display size.
3. **Lock.** Commit everything to source control with a "kit locked" message.

## Use it on Mill Alley

Today is final polish in Camera 1, locking the scene and Review 14. For the standard check, the kit sheet needs dimensions and a scale figure, as Indian street and Le Chlanko show.

# Week 15 · Jan 25 to 31 · Breakdown renders and the write-up

This week's goal is to present the work: breakdown renders that show how it was made, a material-type overlay, a turntable, a short edit, and a written breakdown for ArtStation.

## Mon Jan 25 · Breakdown renders: wireframe, clay and UV checker

**Goal:** render matching wireframe, clay and UV-checker passes of the same frame.

**Time:** about 40 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Create Perfect Clay, Wireframe & Viewport Renders in Blender — Bhagath Goud · 2:42 clay, 4:06 wireframe; 0:35 viewport render (2022, same menus in 5.x)
- **Manual:** Viewport Overlays — Wireframe overlay (opacity/threshold) for the wire passes
- **Manual:** Viewport Shading — Solid mode Lighting: Studio / MatCap for the clay passes; Texture color for UV-checker views (Image > New > Generated Type: UV Grid)
- **Manual:** Lighting (Workbench) — Render the clay pass with the Workbench engine

### Steps

1. **One camera, many passes.** Every breakdown pass uses the exact same camera and frame as the beauty shot, so viewers can compare them or slide between them.
2. **Clay with a material override.** View Layer Properties > *Override* > *Material*: pick a plain grey material. Every object renders with it, without touching your real materials. Clear the field to go back.
3. **Wireframe, quick.** Solid shading, Overlays > *Wireframe* at full opacity, grid and other overlays off, then View > *Viewport Render Image*.
4. **Wireframe, clean.** For crisp lines in a proper render, duplicate the kit into its own collection, add a *Wireframe* modifier (Thickness about 0.005 m) to each copy and give them one dark material. Render that collection on top of the clay pass, as you did for the Week 8 prop sheet.
5. **UV checker.** Make a material with a UV Grid image (Image > New > *UV Grid*) and use it as the override. Even squares everywhere prove consistent texel density.
6. **Save each pass** with a clear name (`Cam01_Clay.png`, `Cam01_Wire.png`, `Cam01_UV.png`).

### Common mistakes

- Passes from slightly different cameras. They won't line up in a slider. Never move the camera between passes.
- A messy wireframe on a busy scene. Show the kit on its own, or a close-up, where the wireframe tells a story.
- Forgetting to clear the override. Your next beauty render comes out grey.

### Drill: wireframe and clay of the kit (40 minutes)

Render clay, wireframe and UV-checker passes of your kit layout from one camera.

**Pass when:** the three passes line up pixel for pixel, the wireframe lines are readable at 1920 px wide, and the UV checker shows even squares across modules.

### Use it on Mill Alley

Today you render the beauty shots at 4K in UE5's Movie Render Queue (Camera 1 plus 4 to 6 angles). Do the Blender passes of the kit layout alongside; UE5's own wireframe and lighting-only view modes give the in-engine versions.

## Tue Jan 26 · The material-type overlay

**Goal:** colour every surface by how it's textured (tiling, trim, unique, decal), the overlay that shows a modular pipeline at a glance.

**Time:** about 30 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Change Material's Viewport Display Color in BLENDER 3D — Brandon 3D · Jan 2024; how to set the per-material colour the overlay uses
- **Manual:** Viewport Shading — Solid > Color: Material / Object / Random / Attribute. Set Material viewport colors (trim = one colour, unique = another)
- **Manual:** Object Color (Workbench) — The same color modes for a final Workbench render

### Steps

1. **Pick a colour per type.** For example: tiling materials blue, trim sheets green, unique textures orange, decals pink, foliage yellow.
2. **Set each material's viewport colour.** Material Properties > *Viewport Display* > *Color*. Set every tiling material's to blue, every trim's to green and so on. This doesn't change renders.
3. **Show it.** Solid shading, Color set to *Material*. Lighting *Flat* gives pure colours; *Studio* keeps some shading so shapes still read.
4. **Render it.** View > *Viewport Render Image* from the same camera as the beauty shot.
5. **Add a legend.** A small key in the corner: blue = tiling, green = trim, orange = unique, pink = decals.
6. **In UE5.** For an in-engine version, make one flat unlit material per colour and temporarily assign it, or use the Blender version of one facade.

### Common mistakes

- Too many categories. Four or five colours read; ten don't.
- No legend. The image means nothing without it.
- Colours too similar. Pick clearly different hues.

### Drill: the trim/unique overlay of one facade (40 minutes)

Make the overlay of the left or right facade with a legend.

**Pass when:** every surface is one of your colours (nothing grey), the legend matches, and the image shows that most of the facade is tiling and trim, as in Lemon.

### Use it on Mill Alley

Today you make the breakdown renders: kit sheet, wireframe, clay, lighting only, material spheres and trim sheets. Add this overlay to that set.

## Fri Jan 29 · Turntables

**Goal:** render a smooth 360° turntable in Blender, as a backup to Marmoset's.

**Time:** about 30 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** Blender Tutorial - How to Make a 360 Turntable Animation — Eve Sculpts · 6:16, watch all (2021; constraints unchanged)
- **Manual:** Track To Constraint — Aim the camera at an Empty on the skybridge
- **Manual:** Follow Path Constraint — Camera on a Bezier circle; animate the offset or the path duration for 10 s (240 frames at 24 fps)
- **Marmoset docs:** Turntable Animation — The main route: Scene > Add Object > Turntable, then parent the mesh or the camera

### Steps

1. **A pivot empty.** Add an Empty (Shift+A > Empty > Plain Axes) at the centre of the asset's base.
2. **Camera on the empty.** Place the camera where you want it, then parent it to the empty: select the camera, Shift-select the empty, Ctrl+P > *Object*. Add a *Track To* constraint on the camera targeting the empty so it always looks at the centre.
3. **Animate the spin.** At frame 1, set the empty's Z rotation to 0 and press I over the field to keyframe it. At the last frame (240 for 10 seconds at 24 fps), set 360 and keyframe again.
4. **Constant speed.** In the Timeline or Graph Editor, select all keys and set Key > *Interpolation Mode* > *Linear* (T in the Graph Editor). Otherwise it eases in and out.
5. **No double frame.** Set the scene's End frame to 240 with the 360° key at 241, so the loop doesn't repeat the first frame.
6. **The path version.** Alternatively: a Bezier Circle around the asset, the camera with a *Follow Path* constraint on it plus Track To the asset. Same result.
7. **Render.** Output Properties: frame rate 24 or 30, output to an image sequence (PNG) or directly to video (Media Type *Video*, see tomorrow). Then Render > *Render Animation* (Ctrl+F12).

### Common mistakes

- A turntable that speeds up and slows down. Interpolation isn't Linear.
- A hitch at the loop point. The first and last frames are the same; end one frame earlier.
- The asset spinning off-centre. Put the empty at the asset's real centre.

### Drill: a 10-second skybridge turntable (40 minutes)

Make a 240-frame turntable of the skybridge with a constant spin, on your studio backdrop.

**Pass when:** it loops with no hitch or speed change, the skybridge stays centred, and it renders out at your portfolio resolution.

### Use it on Mill Alley

Today you render Marmoset turntables for H1, H2 and H6, a prop sheet, and the flythrough sequence. If a Marmoset render fails, this is your backup.

## Sat Jan 30 · The Video Sequencer: editing the flythrough

**Goal:** cut clips together with crossfades and titles and export an H.264 video.

**Time:** about 40 minutes for the lesson and 40 for the drill.

### Watch and read first

- **Video:** A Beginner's Guide to Video Editing in the new Blender 5.0 — Michael Chu · 1:22:15 total (Nov 2025). Watch 6:26-35:37 (adding media, splitting, moving), 35:37 Transitions, 1:02:40 Exporting
- **Manual:** Transitions — Select two overlapping strips > Add > Transition; the overlap length sets the duration
- **Manual:** Cross Strip — The crossfade; Default Fade or keyframed Effect Fader
- **Manual:** Text Strip — Titles
- **Manual:** Output — Output Properties > Output: Media Type = Video; Encoding panel: Container MPEG-4, Video Codec H.264, Output Quality/CRF; Audio subsection

### Steps

1. **A video project.** File > New > *Video Editing*. Set Output Properties > Format: resolution and frame rate to match your renders (e.g. 1920 × 1080 at 30 fps) before adding clips.
2. **Add clips.** Add > *Movie* for video files, or Add > *Image/Sequence* and select all frames of an image sequence. Clips appear as strips on channels.
3. **Cut.** Put the playhead where you want a cut and press K (split). Delete unwanted parts with X. Drag strips to rearrange.
4. **Crossfades.** Overlap the end of one strip with the start of the next on the channel above, select both, then Add > *Transition* > *Cross*. Keep fades short (0.5–1 s).
5. **Titles.** Add > *Text*. Set the text, size and position in the sidebar (N). A title card at the start and your name at the end is enough.
6. **Music (optional).** Add > *Sound*. Use only music you have the right to use; fade it out at the end by keyframing its volume.
7. **Export.** Output Properties > Output: set *Media Type* to *Video*. In the Encoding panel: Container *MPEG-4*, Video Codec *H.264*, Output Quality *High*. Set the frame range to the edit's length, then Render > *Render Animation*.

### Common mistakes

- Clips playing at the wrong speed. The project's frame rate doesn't match the clips; set it first.
- Exporting a PNG sequence by accident. Check Media Type is Video.
- A long, slow edit. 30–45 seconds of strong shots beats two minutes of drifting.

### Drill: a 15-second test edit (40 minutes)

Cut three clips (turntable, a camera move, a clay pass) into a 15-second edit with two crossfades and a title, and export it as H.264 MP4.

**Pass when:** the MP4 plays in a normal video player, it's about 15 seconds, the crossfades are smooth, and the title is readable.

### Use it on Mill Alley

Today you edit the 30 to 45 second flythrough in the Video Sequencer and make the reference vs final side-by-side. The side-by-side is two image strips: in the sidebar, under *Transform*, scale each to half width and move them left and right.

## Sun Jan 31 · The written breakdown

**Goal:** write the ArtStation post text that explains the project like a professional breakdown.

**Time:** about 60 minutes.

## Watch and read first

- **Read:** Presenting an Unreal Engine Environment Art Scene - from a Producer! — CGHero · Jun 2022: thumbnail, hero shot, extra renders, short flythrough, modular kit, material and process breakdowns
- **Read:** Building Victorian Street of Epic Proportions in UE4 — 80 Level (Richard Vinci) · A model written breakdown on a matching subject: Introduction, Meshes, Materials, Scale, Lighting, Challenges
- **Read:** Tips On How To Make Your Portfolio Better — 80 Level (Ellie Harisova) · Tip 4: break the environment down from macro to micro after the beauty shot

## Steps

1. **Intent (2–3 sentences).** What the project is and why: "A 1:1 study of a derelict Victorian mill alley, built in 15 weeks on a AAA modular pipeline to practise kit design and material storytelling."
2. **References.** Credit the reference photo and say how you used it (camera match, scale breakdown).
3. **Kit stats.** Number of modules, trim sheets and tiling materials; module grid and texel density.
4. **Workflow.** One short paragraph per stage: greybox, materials and masks, kit, heroes, foliage, decals and vertex paint, lighting. Mention the tools (Blender, Marmoset, UE5) and what each did.
5. **What you'd do differently.** One honest line. Reviewers like seeing judgement.
6. **Credits.** Every third-party material, HDRI, prop or music track, with its source and licence (ambientCG CC0, Poly Haven CC0, Blendkit, Fab).
7. **Order the images.** Beauty shot first, then alternate angles, then breakdowns (kit sheet, overlays, wireframes, foliage, props), then the video and process shots.

## Common mistakes

- A wall of text. Use short headings and paragraphs.
- No numbers. Kit stats and sizes make the work credible.
- Missing credits. Always credit what you didn't make.

## Use it on Mill Alley

Today you lay out and publish the ArtStation post with this text and update your portfolio. For the standard check, go through all 16 portfolio standards one last time and tick each against your post.